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A Multi-Stage Optimized Adaptive Control Framework for Quadcopter Trajectory Tracking Under Wind Disturbances and Payload Variations

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ABSTRACT

This paper proposes a multi-stage control strategy for quadcopter trajectory tracking under wind disturbances and payload variations. The approach is built on a standard cascaded architecture, where the outer loop regulates horizontal position and generates attitude references, while the inner loop tracks the commanded attitudes. First, Particle Swarm Optimization (PSO) is implemented offline to tune baseline PID controllers for the outer-loop horizontal channels and inner-loop attitude channels using a tracking-based objective function. Then, a supervisory fuzzy self-tuning mechanism is integrated within the inner loop to adapt the attitude PID gains online according to the tracking error and its rate, improving robustness against coupling effects and external disturbances. To ensure effective and bounded adaptation, a Genetic Algorithm (GA) is employed offline to optimize the fuzzy scaling factors. In addition, Linear Active Disturbance Rejection Control (LADRC) is assigned to the altitude channel to explicitly address payload-induced thrust-to-weight mismatch by estimating and compensating for the total disturbance. MATLAB/Simulink simulations demonstrate improved trajectory tracking and disturbance rejection compared with the PSO-tuned fixed-gain baseline. Performance evaluation using the integral of time-weighted absolute error (ITAE) confirms superior robustness under wind disturbance and payload variation scenarios while maintaining bounded control effort.

KEYWORDS:

Quadcopter trajectory tracking, Wind disturbances, Payload variations, Fuzzy-PID; LADRC, Particle swarm optimization

1 INTRODUCTION

In recent years, unmanned aerial vehicles (UAVs), especially quadcopters, have gained high attention due to their expanding range of applications [1–4]. They have been used in wide applications from aerial photography and precision agriculture to search and rescue and infrastructure inspection [3, 5]. The reason for their effectiveness lies in their ability to perform autonomous and precise trajectory tracking [6–9]. However, achieving this precision remains a significant control engineering challenge as it has a non-linear systems behavior and cross-coupling between axes and under actuated configuration [2, 10]. For

example, wind disturbance. Furthermore, quadcopters are highly sensitive to internal uncertainties, such as sudden payload variations, and external disturbance [4, 11, 12]. For example, wind disturbance can affect the stability of the system and degrade its performance [13]. The proportional-integral-derivative (PID) controller still the quadcopter flight control and the useful in tight spaces and missions' industry standard due to its direct implementation and intuitive tuning [2, 10, 14, 15]. However, the fixed-gain nature of conventional PID controllers is the main issue. Although it tuned for adequate performance at a specific operating point, it lacks the adaptability to maintain performance across the entire flight envelope or in the presence of the disturbances and uncertainties [12]. This often results in persistent tracking errors, oscillatory responses, and in some cases, instability [2]. Much research has been conducted in intelligent and adaptive control to address these limitations [10]. For instance, Fuzzy Logic Controller (FLC) offers a model-free approach to handle nonlinear systems and shows highly effective performance [2]. The self-tuning Fuzzy-PID controller, where a fuzzy inference system adjusts the PID gains online based on the tracking error and its derivative (i.e., error and change-rate of error), has demonstrated superior performance over fixed-gain PID in several studies [2, 16–18]. Despite its advantages, the design of the fuzzy system itself including the membership functions and scaling factors often relies on trial-and-error or heuristic methods, which may not yield a globally optimal solution and can be time-consuming [2, 19]. This is where metaheuristic optimization algorithms provide a powerful solution. Particle Swarm Optimization (PSO) and Genetic Algorithms (GA) have been successfully applied to automate and optimize control parameters, avoiding local minima to find high performance solution [10, 20, 21]. While some research has explored using PSO for tuning Fuzzy-PID systems, a gap exists in developing a structured, hierarchical optimization framework that systematically combines the global search capability of these algorithms with the real-time adaptability of fuzzy logic.

This paper proposes a multi-stage quadcopter control framework based on a standard cascaded architecture, where PSO is employed offline to tune the baseline PID controllers of the outer-loop horizontal position channels and the inner-loop attitude channels using a trajectory-tracking objective. In addition, a supervisory fuzzy self-tuning mechanism is integrated within the inner loop to adapt the attitude PID gains online according to the tracking error and its rate, thereby improving robustness against wind-induced coupling and unmodeled dynamics while preserving the simplicity of the PID structure. Moreover, a GA-based optimization is implemented offline to tune the fuzzy scaling factors, ensuring bounded and effective gain adaptation without manual trial-and-error selection. To enhance vertical robustness, LADRC is assigned to the altitude channel with the primary objective of addressing payload variation, where the resulting thrust-to-weight mismatch is treated as a total disturbance and compensated online through disturbance estimation. Finally, comprehensive MATLAB/Simulink simulations under nominal conditions, wind disturbances, and payload variation are conducted, and the results supported by ITAE-based evaluation confirm superior tracking and robustness compared with the PSO-tuned fixed-gain baseline.

The remainder of this paper is organized as follows. Section 2 reviews related work. Section 3 presents the quadcopter model and the proposed multi-stage control design (PSO-tuned outer-loop horizontal and inner-loop PID, inner-loop fuzzy self-tuning with GA-optimized scaling factors, and LADRC altitude control for payload variation), along with the considered disturbance models. Section 4 reports the simulation results under nominal conditions, wind disturbance, and payload variation. Finally, Section 5 concludes the paper and outlines future work.

2 RELATED WORKS

Recent quadcopter trajectory-tracking studies mainly improve robustness through optimized PID-based cascaded structures, intelligent self-tuning, and explicit disturbance rejection. Li et al. [1] designed a cascaded inner–outer controller where the outer position loop uses GA–PSO offline tuning to minimize ITAE, while the inner attitude loop employs LADRC with fuzzy online parameter adaptation. Under time-varying wind with added Gaussian noise, their simulations reported lower ITAE and faster stabilization than PID–PID, PID–LADRC, and PID–Fuzzy LADRC; however, the PID gains remain offline-tuned and validation is largely limited to hovering/waypoint tasks rather than challenging 3D trajectories. Using a Newton–Euler model of a DJI Tello, Baharuddin et al. [2] proposed a Mamdani MIMO self-tuning fuzzy-PID within a standard cascaded position–attitude architecture, updating K_p, K_i, and K_d online from the tracking error and error rate. MATLAB/Simulink results on several reference paths (circle, square, zigzag, spiral, and lemniscate) showed improved IAE/ISE/RMSE and faster tracking compared with fixed-gain PID, though disturbance effects were not considered.

Beyond basic self-tuning, El Hamidi et al. [22] compared conventional PID, neural-based PID gain scheduling, and PSO-optimized fuzzy-PID, showing that improved back-propagation learning and PSO-tuned scaling factors can accelerate convergence and enhance tracking performance. PID tuning has also been framed as a multi-objective optimization problem: Kouritem et al. [12] introduced MONLTA to tune six PID loops (altitude, roll, pitch, yaw, and x–y position) by minimizing metrics such as IAE and ITSE, outperforming GA-based tuning and nonlinear PID (NLPID) in simulation.

For stronger robustness under uncertainty, many works employ nonlinear and sliding-mode–based designs. Madebo [23] presented a Lyapunov-stable neuro-fuzzy adaptive sliding-mode controller derived from a quaternion Newton–Euler model, where an ANN estimates the equivalent control term and fuzzy logic generates the switching term, improving tracking while reducing chattering relative to SMC and fuzzy-SMC. Similarly, Darwito et al. [24] integrated ANFIS with SMC to adapt the switching gain under changing conditions, achieving improved tracking over classical SMC in simulation. Fuzzy approaches have also been validated experimentally: Nguyen et al. [25] proposed a fuzzy-PD trajectory-tracking controller and demonstrated performance gains over conventional PD in both MATLAB/Simulink and real-time experiments on a QDron2 platform.

Several studies explicitly address wind and external disturbances. Li et al. [15] combined fuzzy approximation of unmodeled dynamics with a nonlinear disturbance observer within a command-filtered backstepping framework, reducing derivative complexity and improving experimental precision and stability compared with PID. Along the same direction, Khatoon et al. [26] proposed a hybrid adaptive PD neuro-fuzzy controller that improved transient response, robustness, and disturbance rejection relative to PD, PID, and fuzzy controllers in simulation. From an implementation perspective, Noordin et al. [27] evaluated altitude control on a Parrot Mambo using PID, and adaptive PID (APID) with a fuzzy compensator (APIDFC); the second-order sliding-mode adaptation and fuzzy compensation reduced chattering and power consumption and achieved significantly lower ISE than PID in experiments. For higher-level intelligent robustness, Shirzadeh et al. [28] combined exponential sliding-mode control with a Type-2 fuzzy neural network, optimized via cuckoo search with Lyapunov-based adaptation laws, and reported superior tracking under uncertainties, disturbances, and actuator saturation compared with Type-1 fuzzy neural and GA-tuned methods in simulation.

Overall, Table 1 indicates that existing studies often improve PID tracking without explicit wind evaluation [1], optimize controllers offline without structured online supervision [22], or rely on higher-complexity nonlinear/SMC-based designs for robustness [28]. In contrast, this work retains a PID-compatible cascaded architecture while integrating PSO-based global tuning, fuzzy online gain adaptation with GA-optimized scaling factors, and an LADRC disturbance-rejection layer to compensate lumped uncertainties (e.g., payload variations). This combination provides a systematic, yet practical workflow aimed at robust tracking of disturbed 3D trajectories.

Table 1 Comparison of related works and the proposed approach (PID-, fuzzy-, and optimization-based quadcopter control)

Ref.	Method / Controller core	Online adaptation	Offline optimization	Disturbance handling	Validation	Main strengths	Main limitation / gap (why this work is needed)
[1]	Dual-loop: outer PID + inner LADRC with fuzzy augmentation (cascaded)	Yes (fuzzy adaptation of LADRC parameters)	Yes (GA-PSO tunes PID gains via ITAE)	Yes (time-varying wind + Gaussian noise)	Simulations (mainly hover/step-like references; baseline comparisons)	Lower ITAE; faster stabilization; improved wind/disturbance rejection vs PID/PID–LADRC baselines	Tested on simple trajectories (no complex/aggressive 3D paths)
[22]	Neural-PID vs PID vs PSO-optimized Fuzzy PID	Yes (NN schedules gains)	Yes (PSO tunes scaling factors)	Indirect (improves convergence; disturbance not central)	Simulations	Good comparative framework and improved convergence	Does not focus on wind rejection and does not present a hierarchical tuning + supervision pipeline for cascaded quadcopter control.
[23]	Neuro-fuzzy adaptive SMC (ANN + FLC)	Yes (adaptive estimation)	No (mainly adaptive laws)	Yes (robust; chattering reduced)	Simulations	High robustness; reduced chattering; good tracking	Higher complexity than PID-based designs; harder to tune/implement for practical PID-centered pipelines.
[24]	ANFIS-SMC (ANFIS adapts SMC gain)	Yes (gain adapts online)	No explicit global optimizer	Yes (robust to changes/disturbances)	Simulations	Improved tracking vs classical SMC	SMC-based complexity and chattering considerations; not PID-compatible for easy implementation.
[25]	Fuzzy-PD for quadcopter trajectory tracking	Yes (rule-based fuzzy action)	No explicit global optimizer	Not the main focus	Simulation + real-time experiments	Experimental validation; improved over PD	PD may suffer steady-state errors under persistent disturbances; no explicit wind

							robustness optimization.
[15]	Fuzzy approximation + nonlinear disturbance observer (backstepping + command filter)	Yes (adaptive + observer)	No (model-based design)	Yes (explicit wind/disturbance rejection)	Experiments	Strong disturbance handling and high precision	More complex nonlinear framework; not a PID-based architecture with simple tuning and clear industry compatibility.
[26]	Hybrid adaptive PD neuro-fuzzy controller	Yes (adaptive neuro-fuzzy)	Not emphasized	Yes (targets disturbance rejection)	Simulations	Improved robustness and transient response	Limited details on cascaded PID architecture and systematic global tuning for wind trajectories.
[27]	PID vs APID vs APID + fuzzy compensator (APIDFC)	Yes (online gain adaptation + fuzzy)	No global optimizer	Yes (disturbance rejection; chattering mitigation)	Real experiments	Practical implementation; better ISE; lower power/chattering	Focus on altitude control; does not provide a full 3D trajectory wind study with hierarchical optimization layers.
[28]	ESMC + Type-2 Fuzzy Neural Network + COA optimization	Yes (Lyapunov-based adaptation)	Yes (COA optimizes parameters)	Yes (uncertainty, disturbances, saturation)	Simulations	Very robust and accurate tracking	High complexity; tuning and implementation burden; less aligned with PID-centered practical design.
This work	PSO-tuned cascaded PID (outer-loop horizontal + inner-loop attitude) + inner-loop supervisory fuzzy self-tuning + GA-optimized fuzzy scaling factors + LADRC altitude	Yes (inner-loop fuzzy adapts attitude PID gains online; LADRC/LESO estimates and compensates total disturbance in altitude).	Yes (PSO for baseline PID gains in outer horizontal + inner attitude loops; GA for fuzzy scaling factors).	Yes (explicit evaluation under wind disturbances and payload variation; LADRC provides altitude robustness to payload-induced uncertainty).	MATLAB 3D trajectory simulations (nominal vs wind and payload variation) with ITAE-based comparison	PID-compatible and implementable; systematic baseline tuning + bounded online adaptation; improved wind robustness through adaptive inner loop; payload-robust altitude regulation via LADRC	Validated in simulation; HIL/real-time experimental validation and formal stability analysis of the overall cascaded structure remain as future work.

3 METHODOLOGY

3.1 Quadcopter Modeling

This section presents the nonlinear dynamic model adopted for controller synthesis and simulation. A nonlinear formulation is preferred for aggressive maneuvers and for disturbance studies (e.g., wind), since these effects are not well captured by oversimplified linear models [29, 30].

3.1.1 Reference frames and attitude representation

Two coordinate frames are considered: (i) an inertial/spatial frame $\{S\}$ fixed to the earth and (ii) a body-fixed frame $\{B\}$ attached to the quadcopter at its center of mass. The vehicle position in $\{S\}$ is denoted by $\boldsymbol{\xi}_b^s = [x\ y\ z]^T$, and the attitude is parameterized by the Euler angles (ϕ, θ, ψ), representing roll, pitch, and yaw, respectively (Fig. 1). The rotation matrix $R_b^s(\phi, \theta, \psi)$ maps vectors expressed in $\{B\}$ to $\{S\}$.

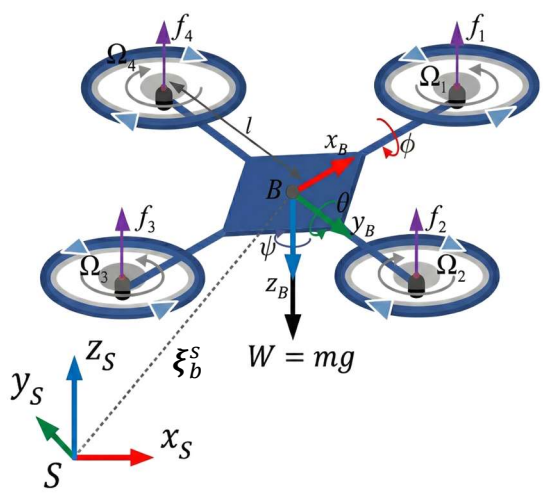


Fig. 1 Inertial and body-fixed reference frames of the quadcopter and the Euler-angle convention [3].

The attitude transformation is constructed via three successive elemental rotations about the body axes. The elemental rotation matrices are given by

$$R_x(\phi) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\phi & -\sin\phi \\ 0 & \sin\phi & \cos\phi \end{bmatrix}, R_y(\theta) = \begin{bmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{bmatrix}, R_z(\psi) = \begin{bmatrix} \cos\psi & -\sin\psi & 0 \\ \sin\psi & \cos\psi & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Accordingly, the body-to-spatial rotation is

$$R_b^s = R_x(\phi)R_y(\theta)R_z(\psi). \tag{1}$$

For compactness, let $c_\alpha = \cos\alpha$ and $s_\alpha = \sin\alpha$. Expanding (1) yields

$$R_b^s = \begin{bmatrix} c_\psi c_\theta & c_\psi s_\theta s_\phi - s_\psi c_\phi & c_\psi s_\theta c_\phi + s_\psi s_\phi \\ s_\psi c_\theta & s_\psi s_\theta s_\phi + c_\psi c_\phi & s_\psi s_\theta c_\phi - c_\psi s_\phi \\ -s_\theta & c_\theta s_\phi & c_\theta c_\phi \end{bmatrix}. \tag{2}$$

The body angular velocity of $\{B\}$ with respect to $\{S\}$, expressed in the body frame, is denoted by $\omega_b^{b,s} = [p\ q\ r]^T$,where p, q, and r are the roll-, pitch-, and yaw-rate components. The Euler-angle rates $[\dot{\phi}\ \dot{\theta}\ \dot{\psi}]^T$ are related to $[p\ q\ r]^T$ through the standard kinematic mapping

$$\begin{bmatrix} \dot{\phi} \\ \theta \\ \dot{\psi} \end{bmatrix} = \begin{bmatrix} 1 & \sin\phi\tan\theta & \cos\phi\tan\theta \\ 0 & \cos\phi & -\sin\phi \\ 0 & \dfrac{\sin\phi}{\cos\theta} & \dfrac{\cos\phi}{\cos\theta} \end{bmatrix} \begin{bmatrix} p \\ q \\ r \end{bmatrix}. \tag{3}$$

This relation is valid for the considered operating envelope (i.e., away from the Euler singularity at $\cos\theta = 0$).

3.1.2 Modeling assumptions

To ensure the model remains both manageable and physically realistic, the following simplifying assumptions are made [31]:

(i) the quadcopter frame is assumed to be rigid; (ii) the structure is assumed to be symmetric; (iii) the center of mass coincides with the origin of the body-fixed frame; (iv) the propellers are treated as rigid components; and (v) thrust and drag forces are assumed to vary proportionally with the square of the propeller speed.

Table 2 summarizes the symbols and parameters used throughout the quadcopter dynamic model and the proposed controller design.

Table 2 Definitions of symbols and parameters used in the quadcopter dynamic model

Parameter	Definition
$\omega_b^{b,s}$	The angular velocity of the body frame with respect to the inertia frame expressed in the body frame

$J^{b,b}$	The inertia of the body expressed in the body frame
m	Quadcopter mass
$f^{b,b}$	The total applied force on the body expressed on the body frame
$\tau^{b,b}$	The total applied torque on the body expressed in the body frame

3.1.3 Newton–Euler dynamics

The quadcopter rigid-body dynamics are derived using the Newton–Euler formulation, yielding coupled rotational and translational equations of motion (Eqs. (4)–(14)). This formulation is widely used for quadrotors and naturally accommodates disturbance terms when required [32, 33].

The rotational motion of the rigid body expressed in $\{B\}$ is given by

$$J^{b,b}\dot{\omega}_b^{b,s} = -\omega_b^{b,s} \times (J^{b,b}\omega_b^{b,s}) + \tau^{b,b}. \quad (4)$$

where $J^{b,b} \in \mathbb{R}^{3\times3}$ is the inertia matrix, $\omega_b^{b,s} = [p\ q\ r]^T$ is the body angular velocity, and $\tau^{b,b}$ is the net control torque about the body axes. The cross-product term $\omega_b^{b,s} \times (J^{b,b}\omega_b^{b,s})$ captures the gyroscopic coupling between rotational axes.

The translational motion of the center of mass in $\{S\}$ is given by

$$m\ddot{\xi}_b^s = R_b^s(t)\, f^{b,b} - mg\hat{z}_s. \quad (5)$$

where m is the quadcopter mass, g is the gravitational acceleration, and $\hat{z}_s = [0\ 0\ 1]^T$ is the unit vector along the inertial z-axis. The vector $f^{b,b}$ is the total applied thrust force expressed in $\{B\}$ (thrust acts along the body z-axis).

The quadcopter is controlled through four virtual inputs: the total thrust u_1 and the body torques (u_2, u_3, u_4). Consistent with the thrust direction, the total thrust force and torque vector are written as

$$f^{b,b} = \begin{bmatrix} 0 \\ 0 \\ u_1 \end{bmatrix}, \tau^{b,b} = \begin{bmatrix} u_2 \\ u_3 \\ u_4 \end{bmatrix}.$$

The total thrust is the sum of individual rotor thrusts,

$$u_1 = k_f(\Omega^2{}_1 + \Omega^2{}_2 + \Omega^2{}_3 + \Omega^2{}_4). \quad (6)$$

where Ω_i is the i-th rotor angular speed and k_f is the thrust coefficient.

The roll, pitch, and yaw torques are produced by differential thrust (lever arm l) and rotor reaction torques (drag coefficient k_m)

$$\begin{bmatrix} u_2 \\ u_3 \\ u_4 \end{bmatrix} = \begin{bmatrix} lk_f({\Omega^2}_4 - {\Omega^2}_2) \\ lk_f({\Omega^2}_3 - {\Omega^2}_1) \\ k_m({\Omega^2}_2 - {\Omega^2}_1 + {\Omega^2}_4 - {\Omega^2}_3) \end{bmatrix}. \quad (7)$$

Equation (7) shows that (i) roll and pitch torques arise from opposing-arm thrust differences, and (ii) yaw torque arises from the imbalance of rotor drag moments due to alternating rotation directions.

Substituting (7) into (4) yields the compact rotational form

$$J^{b,b} \begin{bmatrix} \dot{p} \\ \dot{q} \\ \dot{r} \end{bmatrix} = \begin{bmatrix} lf_4 - lf_2 \\ lf_3 - lf_1 \\ \tau_2 - \tau_1 + \tau_4 - \tau_3 \end{bmatrix} - \begin{bmatrix} p \\ q \\ r \end{bmatrix} \times \left(J^{b,b} \begin{bmatrix} p \\ q \\ r \end{bmatrix} \right). \quad (8)$$

where $f_i = k_f \Omega_i^2$ is the thrust generated by rotor i, and $\tau_i = k_m \Omega_i^2$ is the corresponding reaction (drag) torque.

By substituting $\mathrm{f}^{b,b} = [0\ 0\ u_1]^T$ into (5) and expanding R_b^s, the inertial accelerations become

$$\ddot{x} = \frac{u_1}{m}(\cos\phi\ \sin\theta\ \cos\psi + \sin\phi\ \sin\psi). \quad (9)$$

$$\ddot{y} = \frac{u_1}{m}(\cos\phi\ \sin\theta\ \sin\psi - \sin\phi\ \cos\psi). \quad (10)$$

$$\ddot{z} = -g + \frac{u_1}{m}(\cos\phi\ \cos\theta). \quad (11)$$

Equations (9)– (11) explicitly show that horizontal motion (x, y) is generated by tilting the vehicle (changing ϕ and θ), which redirects thrust into the horizontal plane, while altitude motion depends on the vertical thrust component $\cos\phi \cos\theta$.

Assuming a diagonal inertia matrix

$$J^{b,b} = \begin{bmatrix} J_{xx} & 0 & 0 \\ 0 & J_{yy} & 0 \\ 0 & 0 & J_{zz} \end{bmatrix}.$$

the rotational dynamics can be written explicitly as

(12)

$$\dot{p} = \left(\frac{J_{yy} - J_{xx}}{J_{xx}}\right) qr + \frac{J_r}{J_{xx}} q\,\Omega + \frac{1}{J_{xx}} \tau_\phi.$$

$$\dot{q} = \left(\frac{J_{zz} - J_{xx}}{J_{yy}}\right) pr - \frac{J_r}{J_{yy}} p\Omega + \frac{1}{J_{yy}} \tau_\theta. \quad (13)$$

$$\dot{r} = \left(\frac{J_{xx} - J_{yy}}{J_{zz}}\right) pq + \frac{1}{J_{zz}} \tau_\psi. \quad (14)$$

Here, τ_ϕ, τ_θ, and τ_ψ denote the roll, pitch, and yaw control torques about the body axes, J_r is the rotor rotational inertia, and Ω represents the net rotor-speed term that captures rotor gyroscopic effects according to the rotor rotation directions. The products qr, pr, and pq represent inertial coupling between the rotational axes.

3.1.4 Control inputs and motor mixing (control allocation)

The physical actuators are the four rotor speeds Ω_i, whereas the controller is formulated in terms of the virtual inputs $[u_1\ u_2\ u_3\ u_4]^T$. Therefore, a control-allocation step is required to map the desired thrust and torques to rotor-speed commands. Using thrust and drag relations $f_i = k_f \Omega_i^2$ and $\tau_i = k_m \Omega_i^2$, the motor mixing relationship can be written in matrix form as

$$\begin{bmatrix} \Omega_1^2 \\ \Omega_2^2 \\ \Omega_3^2 \\ \Omega_4^2 \end{bmatrix} = \left(\begin{bmatrix} k_f & k_f & k_f & k_f \\ 0 & -k_f & 0 & k_f \\ -k_f & 0 & k_f & 0 \\ -k_m & k_m & -k_m & k_m \end{bmatrix} \right)^{-1} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \end{bmatrix}. \quad (15)$$

The mixing matrix is determined by the quadcopter geometry and rotor rotation directions [34]. When the matrix is invertible, a unique allocation exists for the commanded thrust–torque vector within actuator limits; in practice, saturation constraints are applied to ensure Ω_i remain feasible.

3.2 Proposed Control Design

This section presents the proposed hierarchically optimized control strategy for quadcopter trajectory tracking. The overall architecture follows a nested inner–outer loop structure, where an outer position loop generates attitude references, and an inner attitude loop ensures fast stabilization and accurate tracking of these references [35]. Such a cascaded design is widely adopted for quadrotor control because it separates the slower translational dynamics from the faster rotational dynamics, improving stability and tuning practicality. Fig. 2 shows the standard nested control loops of quadcopter.

3.2.1 Overview of the control design

The proposed methodology is organized into four consecutive stages to ensure both nominal performance and robustness.

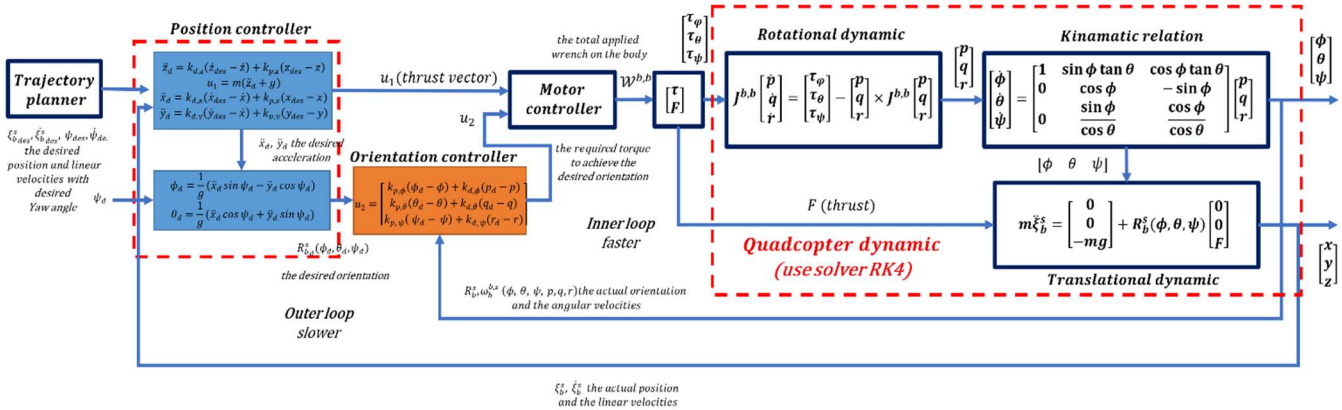


Fig. 2 Cascaded quadcopter control architecture: the outer position loop generating attitude commands for the inner attitude loop.

Stage 1 (offline PSO-tuned baseline PID): Particle Swarm Optimization (PSO) is used to tune the baseline PID gains offline. PSO optimizes (i) the outer-loop horizontal position PID gains (for x–y regulation), which generate attitude references, and (ii) the inner-loop attitude PID gains, which track ϕ_d, θ_d, and ψ_d. This stage establishes strong fixed-gain controllers that serve as a reliable starting point for subsequent robustness layers.

Stage 2 (LADRC altitude loop): The vertical channel is strengthened using Linear Active Disturbance Rejection Control (LADRC). The altitude controller improves robustness to payload variations, which alters the effective mass and introduces thrust-to-weight mismatch. LADRC lumps payload-induced uncertainty, aerodynamic effects, and unmodeled couplings into a single "total disturbance," estimates it online using a Linear Extended State Observer (LESO), and compensates for it directly in the thrust command.

Stage 3 (online supervisory fuzzy self-tuning): Robustness of attitude regulation is enhanced by integrating a supervisory fuzzy self-tuning layer within the inner loop. The fuzzy system uses the attitude tracking error and its rate to generate bounded gain adjustments ΔK_p, ΔK_i, and ΔK_d, thereby adapting the inner-loop PID gains online while preserving the overall cascaded structure.

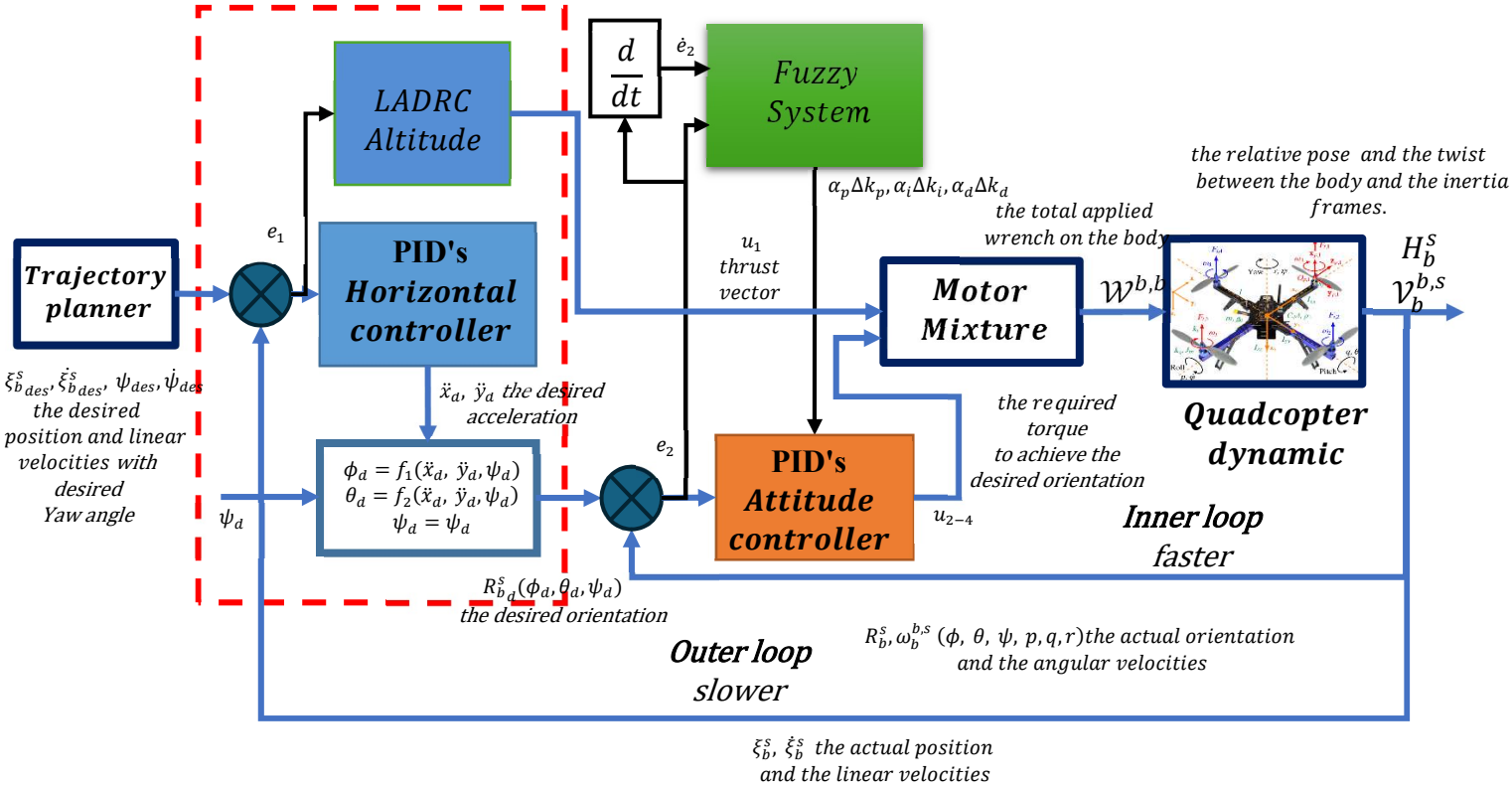


Fig. 4 Integration of the supervisory fuzzy system with the inner PID control loop of the quadcopter and LADRC with the altitude control.

Stage 4 (offline GA optimization of fuzzy scaling factors): A Genetic Algorithm (GA) is used offline to tune the fuzzy scaling factors that govern the amplitude of online gain adaptation. Each candidate scaling-factor set is evaluated via closed-loop simulation, and the scaling factors that yield the best tracking performance are selected while preventing overly aggressive gain variations.

Fig. 4 illustrates the proposed hierarchical cascaded control architecture for the quadcopter, organized into a slower outer loop and a faster inner loop. The trajectory planner provides the desired position/velocity commands and the reference yaw angle (ψ_d). In the outer loop, the horizontal position controller (PID) regulates the xy motion and produces the desired translational accelerations $(\ddot{x}_d, \ddot{y}_d)$, which are mapped together with ψ_d into the desired roll and pitch angles (ϕ_d, θ_d). In parallel, the altitude channel is regulated by LADRC, where the controller and its extended state observer (ESO) estimate and compensate for the lumped disturbance acting on the vertical dynamics (e.g., modeling uncertainties, and payload-induced thrust-to-weight mismatch),

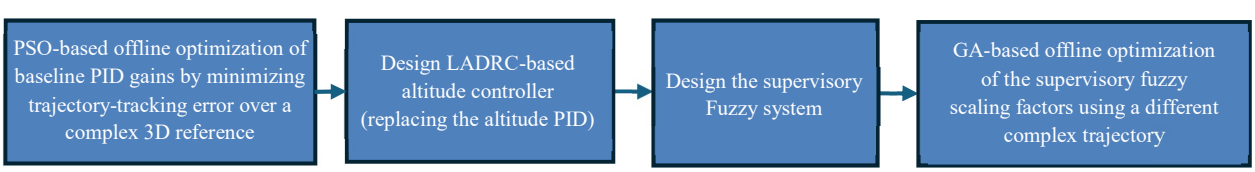


Fig. 3 Hierarchical control-design workflow: PSO-tuned baseline PID, supervisory fuzzy self-tuning, and GA-optimized scaling factors.

yielding a robust thrust-related command for the z-motion. The resulting attitude reference $R_d(\phi_d, \theta_d, \psi_d)$ is then tracked by the inner-loop attitude PID controller, which generates the required body torques. A supervisory fuzzy adaptation layer enhances both the horizontal PID and the attitude PID by updating the gains $(\Delta k_p, \Delta k_i, \Delta k_d)$ online based on the attitude error signals and their derivatives, improving responsiveness and robustness under disturbances. Finally, the motor mixer converts the total thrust u_1 and the body torques (u_2, u_3, u_4) into rotor commands that drive the quadcopter dynamics, closing the feedback loop using measured position, velocity, orientation, and angular rates.

3.2.2 Algorithm 1 - Offline baseline PID synthesis via PSO

In the first stage, baseline PID gains are obtained offline using Particle Swarm Optimization (PSO). The goal of this step is to establish a strong fixed gain starting point for both inner and outer loops before adding online adaptation. As illustrated in Fig. 5, the offline PSO procedure iteratively simulates the quadcopter model and minimizes the tracking-error indices (ITAE/ISE/MSE) to obtain the optimized baseline PID gains (K_{p0}, K_{i0}, K_{d0}). For each controlled channel, the PID law is written as:

$$u(t) = K_p e(t) + K_i \int_0^t e(\tau)\, d\tau + K_d \dot{e}(t) \tag{16}$$

where $e(t)$ is the tracking error.

To ensure that the tuned gains remain effective under strong nonlinear coupling, PSO evaluates candidate gain sets by simulating the quadcopter on a highly exciting reference trajectory (expanding spiral). This trajectory is intentionally selected to persistently excite both outer and inner loops, revealing coupling effects and reducing the risk of "overfitting" the gains to a mild maneuver.

Regarding the cost function a multi-objective function is used to penalize error in complementary ways

$$\begin{aligned} ITAE &= \sum_{k=1}^{N} t_k \cdot \| e(k) \|_1 \cdot \Delta t, \\ ISE &= \sum_{k=1}^{N} \| e(k) \|_2^2 \cdot \Delta t, \\ MSE &= \frac{1}{N} \sum_{k=1}^{N} \| e(k) \|_2^2, \end{aligned}$$

$$J = \alpha \cdot ITAE + \beta \cdot ISE + \gamma \cdot MSE. \qquad (17)$$

Where ITAE emphasizes long-duration residual errors. (encouraging fast settling), ISE penalizes large transient errors (reducing overshoot), and MSE reflects overall tracking accuracy.

PSO iteratively updates particle positions (candidate gains) until convergence, yielding optimized baseline gains $\left(K_{p0}, K_{i0}, K_{d0}\right)$ for use in the next stage.

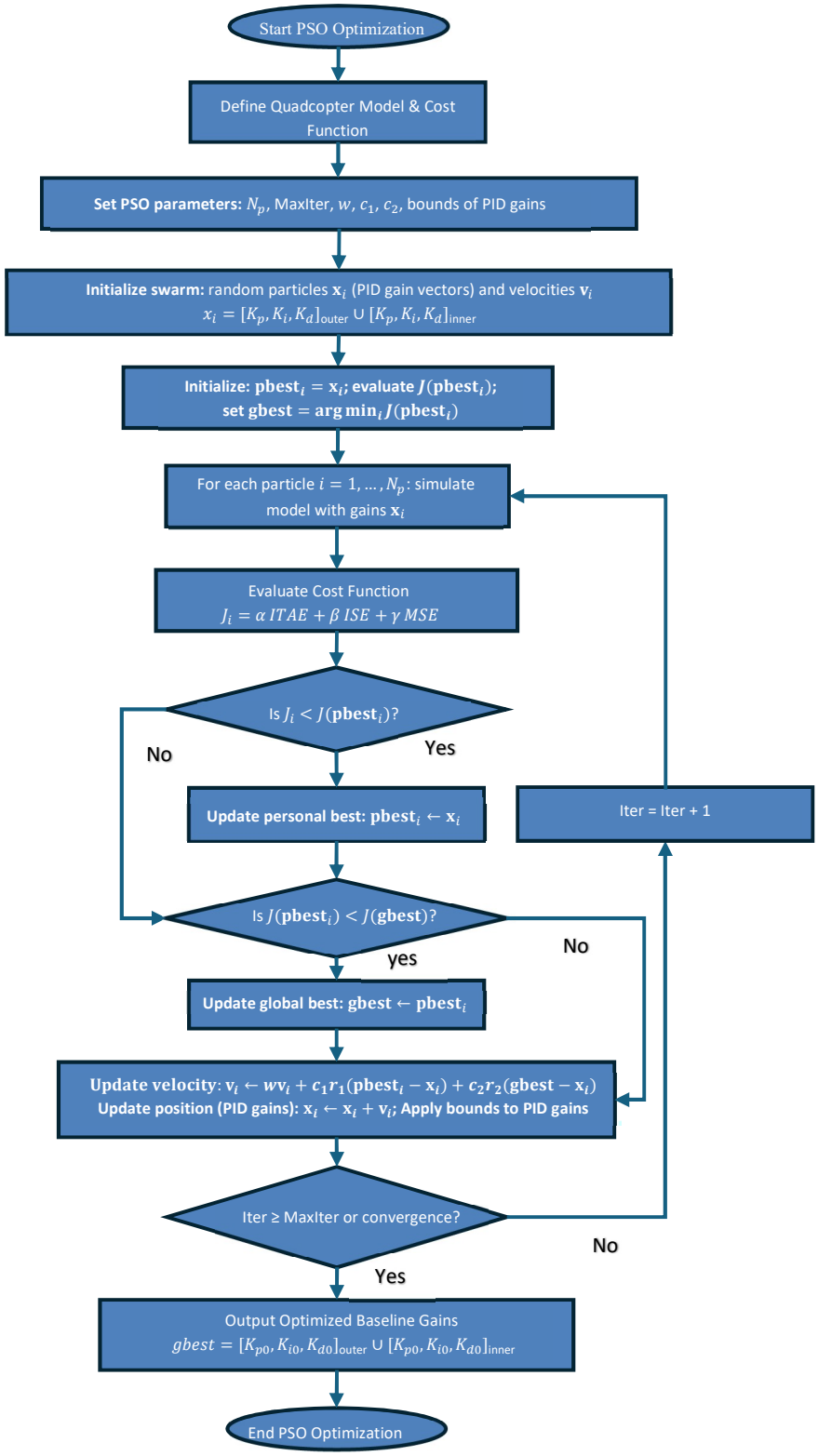


Fig. 5 PSO flowchart (Algorithm 1) for offline tuning of the baseline PID gains .

During the optimization process, quadcopter was given a 3D spiral trajectory with slowly increasing radius to persistently excite both outer and inner loops:

$$\begin{aligned} x_d(t) &= R\big(z(t)\big)\cos(\omega t) \\ y_d(t) &= R\big(z(t)\big)\sin(\omega t) \\ z_d(t) &= v_z t \end{aligned}$$

with radius function

$$R(z) = R_0 + (R_{max} - R_0)\left(\frac{z}{h_{total}}\right)^2. \qquad (18)$$

with ω, ω_z chosen to avoid resonance and actuator saturation. For each controlled axis, PSO searches K_{p0}, K_{i0}, K_{d0} within stability-preserving bounds using a cost.

3.2.3 Design of the LADRC controller for altitude control

To enhance altitude tracking under payload variation, thrust uncertainty, and aerodynamic effects, the altitude loop is designed using LADRC. The key idea is to lump all unknown terms (e.g., mass/thrust mismatch, drag/propwash, and tilt-induced coupling) into a single total disturbance and cancel it online using an observer, instead of relying on an accurate model.

$$\ddot{z} = \frac{\cos\phi\cos\theta}{m} U_1 - g + d_z(t), \qquad (19)$$

where U_1 is the total thrust, m is the mass, and $d_z(t)$ represents unmodeled disturbances affecting the altitude channel (including model mismatch, aerodynamic effects, and payload-induced thrust-to-weight changes). In this work, wind is assumed to have no vertical component, so vertical wind is not included in $d_z(t)$.

$d_z(t)$ is assumed bounded and piecewise continuously differentiable, i.e.,

$$|\, d_z(t) \,| \leq \bar{d}_z, |\, \dot{d}_z(t) \,| \leq \bar{\dot{d}}_z.$$

which is a standard requirement for observer-based disturbance rejection.

Since the attitude loop is significantly faster than the altitude loop, the altitude channel is rewritten in an LADRC-friendly second-order form [36]:

$$\ddot{z} = b_0\, u + f(t), \qquad (20)$$

where $u := U_1$, $b_0(t) = \frac{\cos\phi\cos\theta}{m}$ when tilt compensation is enabled, and $f(t)$ denotes the total disturbance term that includes $-g$ and all uncertainties (including $d_z(t)$).

To estimate and cancel $f(t)$ online, a third-order Linear Extended State Observer (LESO) is used, with states $\hat{z}_1, \hat{z}_2, \hat{z}_3$ estimating $z, \dot{z}, f$, respectively:

$$\begin{cases} \dot{\hat{z}}_1 = \hat{z}_2 + \beta_1 (z - \hat{z}_1) \\ \dot{\hat{z}}_2 = \hat{z}_3 + b_0 u + \beta_2 (z - \hat{z}_1) \\ \dot{\hat{z}}_3 = \beta_3 (z - \hat{z}_1) \end{cases} \tag{21-23}$$

where $\beta_1, \beta_2, \beta_3$ are observer gains. With the disturbance estimate $\hat{z}_3$ the LADRC control law combines a simple PD tracking term with disturbance cancellation:

$$u = \frac{u_0 - \hat{z}_3}{b_0}, u_0 = \ddot{z}_d + k_p (z_d - \hat{z}_1) + k_d (\dot{z}_d - \hat{z}_2), \tag{24}$$

where z_d is the altitude reference, and k_p, k_d set the desired second-order tracking dynamics. The term $-\hat{z}_3/b_0$ actively compensates the lumped disturbance, making the altitude channel behave like a well-damped second-order system while rejecting disturbances online. For tuning, a bandwidth approach is adopted to keep the design repeatable. A controller bandwidth ω_c is selected based on the desired altitude response speed, then $k_p = \omega_c^2$ and $k_d = 2\omega_c$ are used for a non-oscillatory response. Next, an observer bandwidth $\omega_o > \omega_c$ is chosen (faster than the controller), and the LESO gains are set using standard pole placement (e.g., $\beta_1 = 3\omega_o$, $\beta_2 = 3\omega_o^2$, $\beta_3 = \omega_o^3$). Increasing ω_o improves disturbance rejection but can amplify noise measurement, so it is selected according to the sensor quality. Finally, the thrust command is saturated within actuator limits, and practical safeguards (e.g., limiting the disturbance estimate and avoiding aggressive steps) are applied to maintain robustness.

Finally, several practical points are enforced to ensure robust operation on the quadcopter. Gravity can be handled either as part of $f(t)$ or via feedforward (adding mg) to reduce the observer burden. Tilt compensation is recommended during aggressive maneuvers by updating $b_0(t) = \frac{\cos\phi\cos\theta}{m}$, which prevents altitude loss when the vehicle tilts. In addition, thrust saturation is applied at the actuator level, and the LESO must use the saturated thrust value (the actually applied input) to

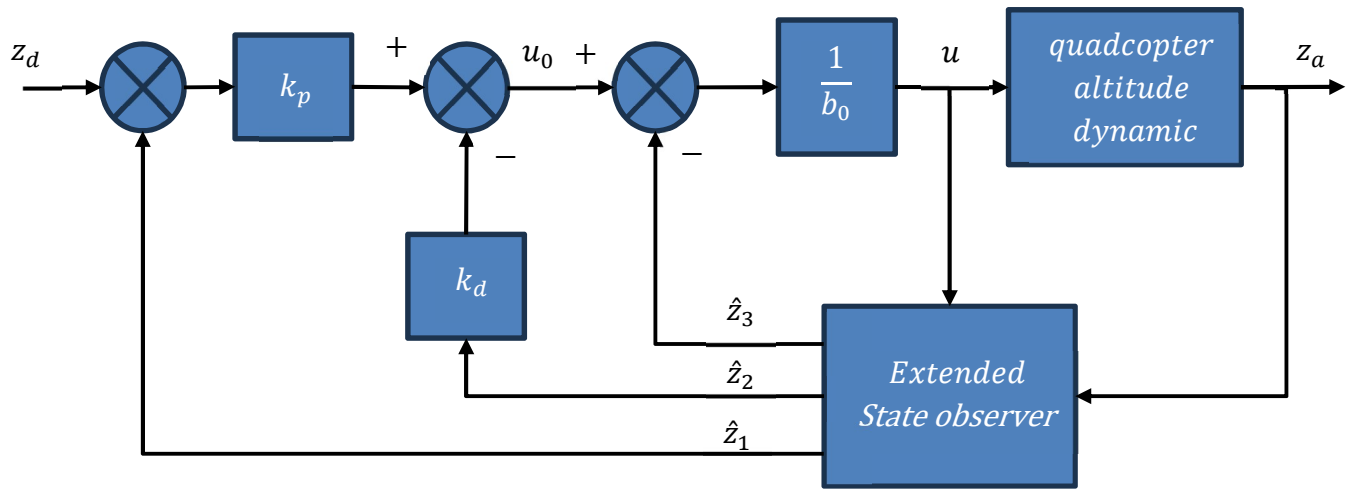


Fig. 6 LADRC-based altitude control structure with ESO estimation and online disturbance compensation [35].

avoid observer windup and biased disturbance estimation. Fig. 6 illustrates LADRC-based altitude control structure with ESO estimation and online disturbance compensation.

3.2.4 Supervisory fuzzy self-tuning layer (online adaptation)

Although the PSO-tuned PID provides good nominal performance, fixed gains can degrade under wind, and unmodeled dynamics. Therefore, a supervisory fuzzy self-tuning mechanism is added to adapt the PID gains online using error information as shown in Fig. 7. The fuzzy system takes as inputs the normalized tracking error and its rate $e_n \in [-1,1]$, $\dot{e}_n \in [-1,1]$ and produces gain adjustments $\Delta K_p, \Delta K_i, \Delta K_d$ where the updated gains are computed by:

$$K_p = K_{p0} + \alpha_p \, \Delta K_p, \qquad K_i = K_{i0} + \alpha_i \, \Delta K_i, \qquad K_d = K_{d0} + \alpha_d \, \Delta K_d$$

where $\alpha_p, \alpha_i, \alpha_d$ are scaling factors that shape the aggressiveness of adaptation.

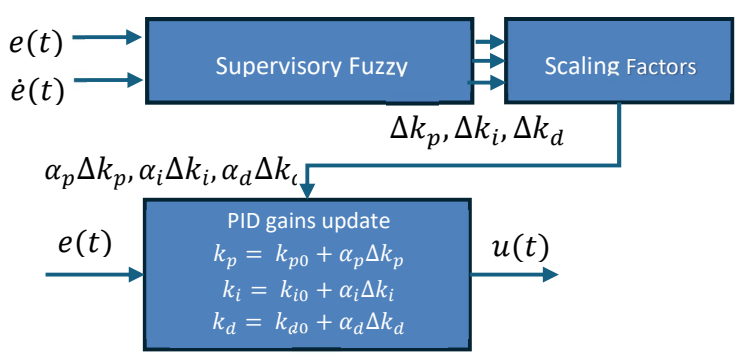


Fig. 7 Supervisory fuzzy self-tuning PID scheme with scaling factors updating ΔK_p, ΔK_i, and ΔK_d.

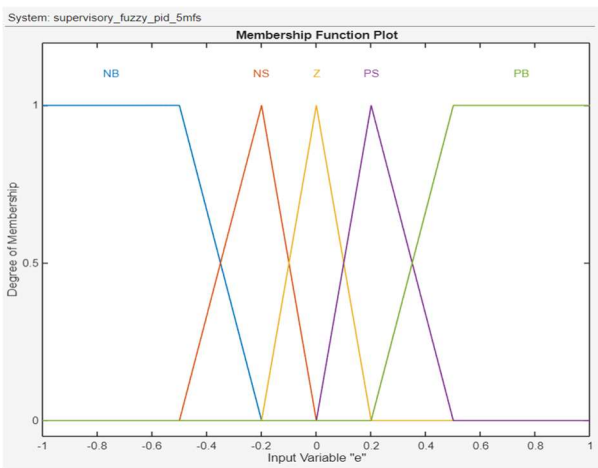
A compact Mamdani-style design is adopted where five triangular membership functions (NB, NS, Z, PS, PB) are assigned to each input and output, and a 5×5 rule base (25 rules) is constructed from

standard PID tuning intuition (increase K_p when error is large, increase K_d when oscillations appear, limit K_i when overshoot risk is high). The inference and aggregation follow min–max logic, and the outputs are defuzzified using the centroid method.

This online tuning philosophy (updating K_p, K_i, K_d based on error and error rate within a cascaded quadrotor structure) consists of prior fuzzy-based quadrotor controllers that report improved tracking versus fixed PID.

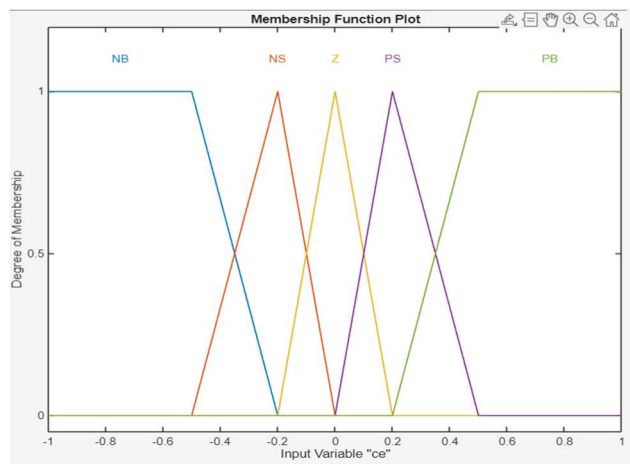
The input membership functions for the normalized attitude error (e) is shown in Fig. 8, which uses five triangular membership functions {NB, NS, Z, PS, PB} showing the normalized range [-1, 1]. This provides comprehensive coverage of all possible error conditions from strongly negative to strongly positive.

Where the input membership functions for the normalized change in error ($\dot{e}$) is illustrated in Fig. 9 with the same membership functions of the error (NB, NS, Z, PS, PB) it is distributed uniformly across the normalized range [-1, 1], This ensures balanced sensitivity to both positive and negative rate errors in the fuzzy inference process.



(a)

Fig. 8 Input membership functions for the normalized tracking error e.



(b)

Fig. 9 Input membership functions for the normalized change of error $\dot{e}$.

Fig. 10 depicts the output membership functions for the proportional gain adjustment (Δk_p) which has the same membership functions of the input (NB, NS, Z, PS, PB). Where the distribution of these membership functions ensures rapid response for tracking errors.

Similar to the membership functions of the proportional gain adjustment (ΔK_p), the output membership functions for the integral gain adjustment (Δk_i), and derivative gain adjustment (Δk_d), utilizing five triangular functions (NB, NS, Z, PS, PB) with a shorter range of [-0.2, 0.2] as shown in Fig. 11 and 12 .this is the common way of the adjustment approach employed for both integral and derivative gains to prevent windup issues and noise amplification.

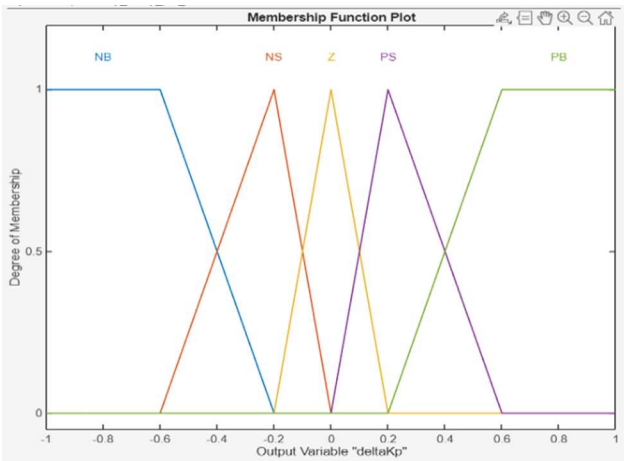


Fig. 10 Output membership functions for proportional gain adjustment ΔK_p.

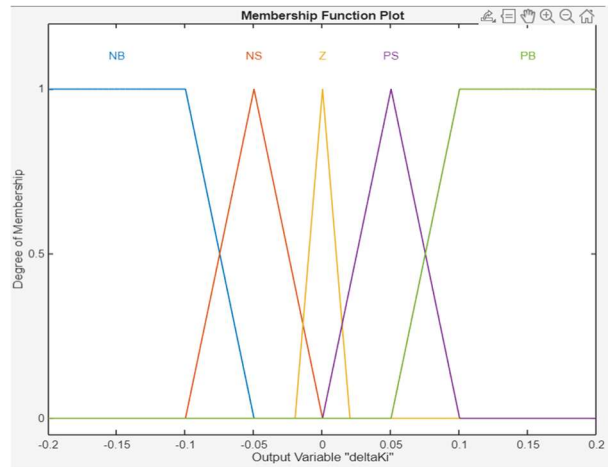


Fig. 11 Output membership functions for integral gain adjustment ΔK_i.

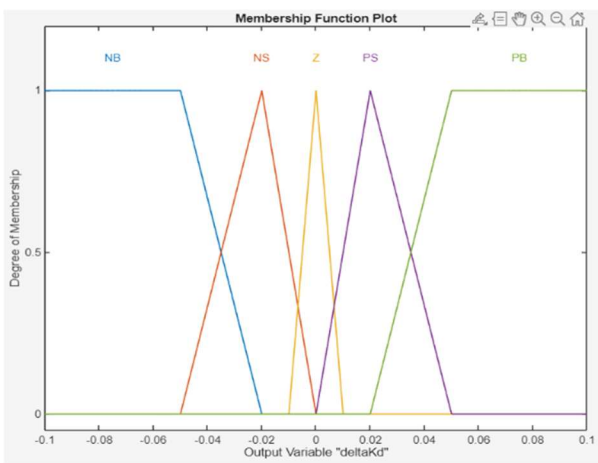


Fig. 12 Output membership functions for derivative gain adjustment ΔK_d

The following tables 3-5 illustrates a 5×5 fuzzy-rule tables for the supervisory fuzzy system. Each cell gives the linguistic output (NB, NS, Z, PS, PB) (negative/positive big or small, or zero). Where Table 3 maps $(e, \dot{e})$ to the change in proportional gain ΔK_p, Table 4 to ΔK_i, and Table 5 to ΔK_d.

For any pair $(e, \dot{e})$, the selected label indicates whether to strongly or weakly increase, decrease, or keep the corresponding gain near zero.

Table 3 Fuzzy rule base for proportional gains update ΔK_p (inputs: e, $\dot{e}$).

e \ $\dot{e}$	NB	NS	Z	PS	PB
NB	NB	NB	NS	NS	Z
NS	NB	NS	NS	Z	PS
Z	NS	Z	Z	Z	PS
PS	NS	Z	PS	PS	PB
PB	Z	PS	PS	PB	PB

Table 4 Fuzzy rule base for integral gains update ΔK_i (inputs: e, $\dot{e}$).

e \ $\dot{e}$	NB	NS	Z	PS	PB
NB	PB	PS	PS	PS	PB
NS	Z	PS	PS	Z	NS
Z	Z	PS	Z	PS	Z
PS	NS	Z	PS	PS	Z
PB	NB	NB	NS	NS	NB

Table 5 Fuzzy rule base for derivative gain update ΔK_d(inputs: e, $\dot{e}$).

e \ $\dot{e}$	NB	NS	Z	PS	PB
NB	PB	PS	Z	NS	NB
NS	PB	PS	Z	NS	NB
Z	NS	Z	Z	Z	PS
PS	NB	NS	Z	PS	PB
PB	NS	NS	Z	PS	PB

3.2.5 Algorithm2: Offline optimization of the scaling factors via (GA)

After obtaining the baseline PID gains (K_{p0}, K_{i0}, K_{d0}) from the PSO stage, this step optimizes the scaling factors of the supervisory fuzzy self-tuning layer offline. In our implementation, each GA individual (chromosome) is defined as a 3-dimensional vector:

$$\alpha = [\alpha_p, \alpha_i, \alpha_d]$$

where $\alpha_p, \alpha_i, \alpha_d$ regulate how aggressively the fuzzy outputs $\Delta K_p, \Delta K_i, \Delta K_d$ modify the baseline gains. For a candidate α, the quadcopter is simulated along the same evaluation scenario used in the paper, while the online gains are updated according to

$$K_p = K_{p0} + \alpha_p \, \Delta K_p, K_i = K_{i0} + \alpha_i \, \Delta K_i, K_d = K_{d0} + \alpha_d \, \Delta K_d.$$

The fitness is computed as J, where J is the tracking cost defined in Eq. (23) (based on the error measures used in the paper, e.g., (ITAE/ISE/MSE). Therefore, each individual is evaluated by

running a full nonlinear simulation and computing J, and the GA searches for the scaling factors that minimize this cost.

Algorithm 2 summarizes the adopted workflow. The GA is implemented with population size = 30, maximum generations = 20, crossover probability = 0.8, mutation probability = 0.2, and elitism = 2 (best individuals are copied to the next generation). Tournament selection is used to form parents, Simulated Binary Crossover (SBX) is applied to generate offspring, and Polynomial Mutation is used to preserve diversity. The best solution at termination is selected and then fixed for all subsequent simulations to keep the comparisons consistent.

ALGORITHM 2: Genetic Algorithm workflow was used to optimize the fuzzy scaling factors.

Input: Population size N, Maximum generations G, Crossover rate Pc, Mutation rate Pm
Output: Optimized scaling factors $[\alpha_p, \alpha_i, \alpha_d]$
1: **Initialize:**
- Population $P \leftarrow$ Generate random population of N individuals
- Each individual = $[\alpha_p, \alpha_i, \alpha_d] \in [0, 30]^3$ (scaling factors bounds)
- Fitness function $F(individual) = 1 / (1 + J)$ where J is cost from Eq. (17)

2: for $generation = 1$ to G do
3: Evaluate fitness for all individuals in P using quadcopter simulation
4: **Selection:**
$Parents \leftarrow Tournament\ selection\ (P, tournament_size = 3)$
5: **Crossover:**
for each parent pair in Parents do
if $rand() < P_c$ then
$Offspring \leftarrow Simulated\ Binary\ Crossover(parent1, parent2)$
else
$Offspring \leftarrow parent1, parent2$
end if
end for
6: **Mutation:**
for each offspring do
if $rand() < P_m\ then$
$offspring \leftarrow Polynomial\ Mutation(offspring, \eta m = 20)$
end if
end for
7: Replacement: $P \leftarrow (Offspring \cup Elitism(P, top\ 10\%))$
8: end for
9: **Return best individual** $[\alpha_p, \alpha_i, \alpha_d]$ from final population

3.3 Disturbance model

To evaluate controller robustness under realistic operating conditions, the quadcopter translational dynamics are augmented with external wind disturbance. The translational dynamics can be written as

$$m\ddot{\xi}_b^s = R_b^s f^{b,b} - mg\hat{\boldsymbol{z}}_s + \mathrm{F}_w(t). \tag{25}$$

where $\ddot{\xi}_b^s = [x\ y\ z]^T$, $\hat{\boldsymbol{z}}_s = [0\ 0\ 1]^T$, and $F_w(t) = \begin{bmatrix} F_x(t) \\ F_y(t) \\ F_z(t) \end{bmatrix}$

denotes the wind-induced force disturbance expressed in the inertial frame $\{S\}$.

3.3.1 Wind disturbance model

In this work, wind is modeled as the superposition of a deterministic horizontal gust and a stochastic turbulence term

$$\mathrm{F}_w(t) = \mathrm{F}_{\text{gust}}(t) + \mathrm{F}_{\text{turb}}(t). \tag{26}$$

This structure follows common practice in quadrotor wind-environment modeling where the wind profile is decomposed into deterministic and stochastic components and deterministic vertical wind is often neglected for tractable analysis and simulation.

1) Gust component (deterministic, time-limited)

$$\mathrm{F}_{\text{gust}}(t) = \begin{cases} A_g\, mg\, \mathrm{d}_{xy}, & t \in [t_g,\, t_g + T_g], \\ 0, & \text{otherwise}, \end{cases}. \tag{27}$$

where A_g is a dimensionless amplitude, t_g and T_g define the gust start time and duration, and d_{xy} is a fixed horizontal direction vector (no deterministic vertical gust is applied).

2) Turbulence component (stochastic)

$$\mathrm{F}_{\text{turb}}(t) = A_t\, mg\, [w_x \quad w_y \quad 0]^T, w_x, w_y \sim \mathcal{N}(0,1). \tag{28}$$

where A_t controls turbulence intensity and the vertical component is set to zero. This is consistent with wind models that represent the stochastic component using Gaussian processes, for example, Bae and Kang [37] explicitly define the wind profile as a sum of deterministic and stochastic terms, noting that the stochastic wind follows a white Gaussian distribution (with Dryden spectral shaping in their study).

3.3.2 Payload-drop disturbance (abrupt mass variation)

A payload-drop is modeled as an instantaneous switch in the total mass at time t_d

$$m(t) = \begin{cases} m_0 + m_p, & t < t_d, \\ m_0, & t \geq t_d, \end{cases} \quad (29)$$

where m_0 is the nominal quadcopter mass and m_p is the payload mass. This captures the sudden change in thrust-to-weight ratio that occurs in "mass-drop" missions and is known to degrade altitude tracking performance if the controller is not sufficiently robust or adaptive. In the proposed framework, this effect is explicitly handled by the LADRC altitude loop through online estimation and compensation of the lumped disturbance induced by the abrupt mass change.

4 SIMULATION RESULTS

To validate the proposed multi-stage control design, a set of MATLAB/Simulink simulations was conducted to evaluate tracking accuracy, robustness, and control effort under both nominal and disturbed conditions. The baseline controller is the PSO-tuned cascaded PID, where PSO optimizes the outer-loop horizontal PID controllers and the inner-loop attitude PID controllers. Then, robustness is enhanced by integrating a supervisory fuzzy self-tuning system within the inner loop, while the altitude channel is strengthened using LADRC to handle payload variations that cause mass/thrust mismatch.

4.1 Simulation Setup Parameters

The quadcopter physical and aerodynamic parameters used in the simulations are listed in the corresponding parameter Table 6. Using the same plant model for all controllers ensures that performance differences are due to the control strategies rather than modeling changes.

Table 6 Quadcopter physical parameters used in the simulation model [3].

Parameters	Description	Value (units)
m	Mass	$1.12\ kg$
I_{xx}	Moment of inertia (roll)	$0.0119 kg.m^2$
I_{yy}	Moment of inertia (pitch)	$0.0119\ kg.m^2$
I_{zz}	Moment of inertia (yaw)	$0.0223\ kg.m^2$
l	Arm length	$0.23\ m$
b	Thrust coefficient in $f = b\,\Omega^2$	7.73×10^{-6}
d	Drag (torque) coefficient in $\tau = d\,\Omega^2$	$1.275 \times 10^{-7} Nms^2$

J_r	Rotor inertia	$8.5 \times 10^{-4} kg.m^2$
g	Gravity	$9.81\ m \cdot s^{-2}$

Next, the reference trajectory is defined using an expanding 3D spiral, where the radius increases with time (time-varying $r(t)$) to excite both translational and rotational dynamics and to reveal coupling effects during motion. Importantly, the same reference trajectory is applied to all compared controllers to guarantee a fair assessment. Table 7 illustrates spiral reference trajectory parameters which are related to Eq 18.

Table 7 Spiral reference trajectory parameters used for excitation and tuning.

Item	Symbol / code variable	Value	Notes / definition
Initial radius	R_0	1 m	Spiral radius at $z = 0$
Maximum radius	$R_{\max}$	3.0 m	Radius at $z = h_{total}$
Total height	h_{total}	8.0 m	Final altitude at the end of the spiral
Angular speed	ω_{total}	2.5 rad/s	Angular velocity in the xy-plane

After defining the model and trajectory, the optimization and adaptation settings are specified. The PSO configuration table reports the swarm and iteration settings used to obtain the baseline PID gains. Table 8 shows the PSO configuration used for offline optimization of PID gains

Table 8 PSO configuration used for offline optimization of PID gains.

Item	Value	Notes
Swarm size	30	Number of particles
Max iterations	30	Optimization iterations
Inertia weight	0.1	Velocity inertia term
Cognitive coefficient	1.5	Pull toward personal best
Social coefficient	1.5	Pull toward global best

In addition to PSO, the GA configuration table reports the genetic parameters used to optimize the fuzzy scaling factors that regulate the strength of online gain adaptation inside the inner loop. Finally, the quantitative comparison is summarized using the ITAE metric, which reflects tracking performance while emphasizing persistent error over time. Table 9 depicts the GA configuration used for offline optimization of fuzzy scaling factors.

Table 9 GA configuration used for offline optimization of fuzzy scaling factors.

Item	Value	Notes
Population size	30	Individuals per generation
Max generations	20	Evolution iterations
Crossover probability	0.8	Applied per mating pair
Mutation probability	0.2	Applied per gene (dimension-wise)
Elitism	2	Best individuals copied to next generation

Table 10 The parameters of the disturbance model

Model part	Symbol or Expression	Value
Payload-drop time	t_d	$t_d = 8$ s
Initial mass	$m = m_0 + m_p$	$m = 1.12 + 0.5 = 1.62$ kg
Post-drop mass	$m = m_0$	$m = 1.12$ kg
Gust window	$t_g, \Delta t_g$	$t_g = 15$ s, $\Delta t_g = 2$ s
Gust force F_{gust} amplitude	A_g	1.0
Turbulence force F_{turb} amplitued	A_t	$A_t = 0.05$
Gust force vector	d_{xy}	$\begin{bmatrix} 1 \\ 0.5 \\ 0 \end{bmatrix}$

To make the robustness tests reproducible and to ensure a fair comparison between controllers, the external inputs used in the disturbance scenarios are fully specified and kept identical across all simulations. Table 10 summarizes the complete disturbance-model parameters used in this study, including the gust activation window, the turbulence contribution, and the payload-drop configuration (initial/post-drop mass and switching time). These parameters define the exact wind/payload profiles injected into the translational dynamics and are therefore the basis for interpreting the tracking deviations and control-effort changes reported in the subsequent results.

Table 11 LADRC parameters

Loop	Symbol	Value (rad/s)
Controller bandwidth	ω_c	3.0
ESO (observer) bandwidth	ω_o	15.0

For the vertical channel, LADRC tuning is expressed compactly in terms of the controller bandwidth and the observer (ESO) bandwidth. Table 11 lists the selected bandwidths used to compute the LADRC/LESO gains for altitude regulation, where the observer bandwidth is chosen higher than the controller bandwidth to enable fast estimation and compensation of the lumped disturbance (e.g., thrust-to-weight mismatch due to payload variation). Fixing these LADRC parameters across scenarios isolates the effect of the proposed multi-stage framework and supports consistent comparison under nominal and payload-varying conditions.

4.2 Offline PSO Tuning of Cascaded PID Gains

This subsection reports the outcome of the offline PSO tuning stage. Fig. 13 illustrates the PSO convergence behavior by plotting the best cost value (global best) versus iteration. The decreasing trend indicates that the swarm progressively improves the candidate solutions until it reaches a stable optimum, confirming convergence of the optimization process. For clarity in the manuscript, the y-axis should be interpreted as the best objective value, which represents the best-performing particle at each iteration.

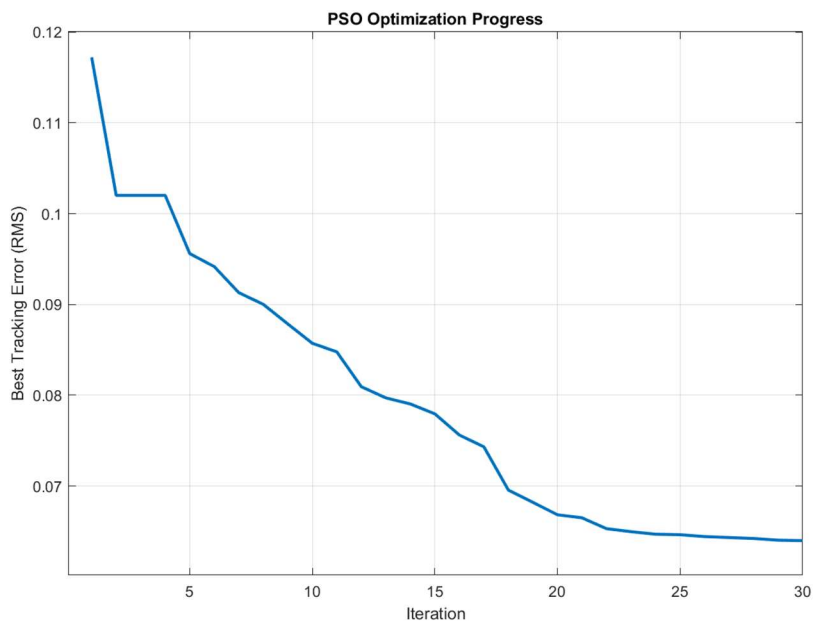


Fig. 13 PSO optimization progress (convergence) for the baseline PID parameters

The resulting optimized gains are then reported in Table 12. These gains are used as the baseline controller for the nominal and disturbance tests presented next.

Table 12 Optimized outer-loop (position) and inner loop PID gains obtained via PSO.

Axis	K_p	K_i	K_d
X	16.90	1.300	6.50
Y	8.29	0.877	2.68
Z	-	-	-
Roll (ϕ)	18.20	0.48	0.75
Pitch (θ)	11.58	0.75	2.00
Yaw (ψ)	6.83	0.53	1.06

4.3 Nominal Trajectory Tracking with the PSO-Tuned PID (Disturbance-Free).

After obtaining the baseline gains, the controller is first evaluated in a disturbance-free scenario to establish a reference level of performance. Under nominal conditions, the PSO-tuned cascaded PID achieves stable closed-loop behavior and successfully tracks the expanding spiral trajectory with bounded errors. The trajectory plot confirms that the vehicle follows the reference path without divergence, while the corresponding attitude and position responses show well-damped behavior and acceptable transient characteristics.

The nominal 3D trajectory tracking obtained with the PSO optimized PID is shown in Fig. 14, and the corresponding attitude/position time responses and control effort (total thrust and body torques) are reported in Figs. 20 and 24, respectively.

This nominal case is essential because it verifies that the PSO tuning yields a feasible and efficient baseline before introducing robustness layers. The next subsection then stresses the same controller under disturbances to reveal its limitations.

4.4 Disturbance Sensitivity of the PSO-Tuned PID Controller.

The key robustness test is performed by applying external disturbances and uncertainty effects during tracking. Although the baseline PSO-PID performs well in the nominal case, the disturbed scenario reveals a clear limitation: tracking accuracy deteriorates and the controller demands higher control effort to compensate for unmodeled effects. This degradation appears as larger deviations from the reference trajectory, increased oscillations in the state responses, and a noticeable increase in the control inputs.

Under the applied horizontal wind disturbance, the resulting deviation in 3D trajectory tracking is visualized in Fig. 15, while the associated attitude/position responses and the increase in control inputs are presented in Figs. 21 and 25, highlighting the limitation of fixed-gain tuning in disturbed

The elevated control activity under disturbance is not only a performance issue but also a practical concern, as it may push actuators closer to saturation and potentially accelerate wear. These observations motivate the need for an adaptive inner-loop enhancement, which is introduced in the next subsection through the supervisory fuzzy self-tuning mechanism.

4.5 Nominal Tracking Improvement Using Fuzzy Self-Tuned Attitude Control.

To improve performance beyond the fixed-gain baseline, the supervisory fuzzy self-tuning layer is integrated inside the inner-loop attitude controller. In the nominal case, this addition improves the closed-loop response by enabling the inner loop to adjust its gains according to the instantaneous tracking error and error rate. As a result, the attitude tracking becomes tighter, which directly improves horizontal tracking because the outer-loop commands are executed more accurately by the inner loop.

Compared with the baseline PSO-PID, the fuzzy-augmented inner loop yields smaller deviations, improved transient regulation, and smoother compensation behavior. Importantly, this improvement is achieved without changing the overall cascaded structure; the enhancement is introduced through bounded online gain adaptation within the inner loop.

For the disturbance-free case with the supervisory fuzzy self-tuning inner loop, the improved 3D trajectory tracking is shown in Fig. 16, with the corresponding attitude/position responses given in Fig. 22 and the resulting control inputs in Fig. 26.

4.6 Superior Disturbance Rejection of the Fuzzy-PID Controller

The disturbance-rejection capability of the proposed controller is evaluated by introducing time-varying wind forces in the horizontal axes, as shown in Fig. 31, which provides the disturbance signals applied in the x and y directions. Under this wind profile, the 3D trajectory in Fig. 17 shows that the GA-optimized fuzzy self-tuning PID maintains close tracking of the reference path, while avoiding the large deviations observed with the fixed-gain PSO-PID baseline under disturbance.

This improvement is further clarified by the time-domain responses in Fig. 23, where the proposed controller achieves faster disturbance attenuation and smaller oscillations in the coupled channels, reflecting stronger closed-loop damping during wind excitation. In addition, the control effort in Fig. 27 remains bounded and structured, indicating that the controller rejects wind disturbances

through effective compensation rather than excessive high-frequency activity. Overall, the observed behavior is consistent with the controller architecture: the inner-loop fuzzy self-tuning adapts the attitude response online, improving the execution of outer-loop commands during wind-induced coupling, while the altitude loop remains robust due to the disturbance-rejection mechanism.

4.7 PSO optimized PID under payload variation

To evaluate robustness against payload variation, the quadcopter mass is switched abruptly according to the payload-drop model defined in Section 3.3.2. This change directly alters the thrust-to-weight ratio and disturbs the vertical channel, which then propagates to horizontal motion through attitude coupling. As a result, even though the PSO-PID controller remains well-behaved in the nominal case, its fixed gains cannot fully accommodate the sudden change in effective plant dynamics, leading to larger transient errors and slower recovery.

This limitation is evident in the ITAE results in Table 13. Compared to the disturbance-free case, the baseline PSO-PID shows a clear increase in time-weighted error under payload variation, particularly in the channels most affected by coupling and required re-balancing after the mass change. This indicates that the fixed-gain baseline is sensitive to abrupt parameter variation, and motivates the use of an explicit robustness mechanism (disturbance estimation/cancellation in altitude) and adaptive inner-loop regulation.

The impact of the payload change on the nominal (PSO-PID + LADRC) controller is illustrated by the 3D trajectory tracking in Fig. 18 (with the corresponding thrust/torque commands shown in Fig. 28), where the transient degradation becomes clear immediately after the mass variation.

4.8 PSO-optimized PID with a self-tuning fuzzy under payload variation

In this payload-variation scenario, the main purpose of integrating LADRC in the altitude channel is to explicitly address the mass/thrust mismatch caused by payload change. Instead of relying on a fixed model, LADRC treats the payload-induced change as part of the total disturbance, estimates it online via the observer, and compensates it directly in the altitude control action. This helps the quadcopter maintain vertical regulation after the mass switch, which also reduces the coupling-driven degradation in horizontal tracking.

The quantitative results in Table 13 support this behavior: under payload variation, the proposed controller achieves slightly lower ITAE than the baseline in key channels (e.g., improvements in

x, y, and z), indicating better recovery after the mass change, while maintaining comparable performance in the remaining channels. Minor trade-offs may still appear in some attitude channels because the fuzzy inner loop prioritizes maintaining stability and bounded control under sudden dynamics change, but overall the controller preserves stable tracking without divergence and with consistent closed-loop behavior under payload variation.

The imposed mass variation used in this scenario is summarized in Fig. 29, and the resulting 3D trajectory tracking and control inputs of the proposed controller are presented in Figs. 19 and 30, respectively, demonstrating stable recovery after the payload switch.

Finally, the qualitative improvement is supported quantitatively by the ITAE comparison: in Table 13, the wind-disturbance case leads to unbounded (inf) ITAE for the PSO-PID baseline (indicating loss of tracking / divergence in the simulation), whereas the proposed method achieves finite ITAE values across all controlled states, confirming stable tracking under wind excitation and payload variations.

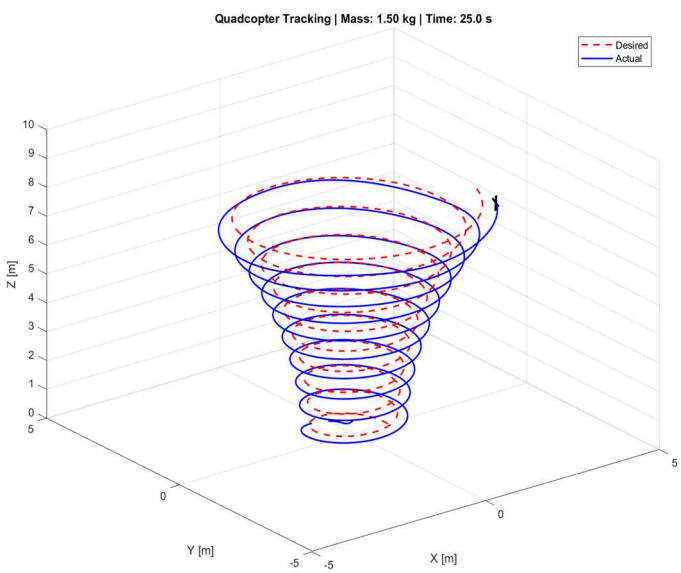


Fig. 14 3D trajectory tracking using the PSO-optimized PID controller (no external disturbance).

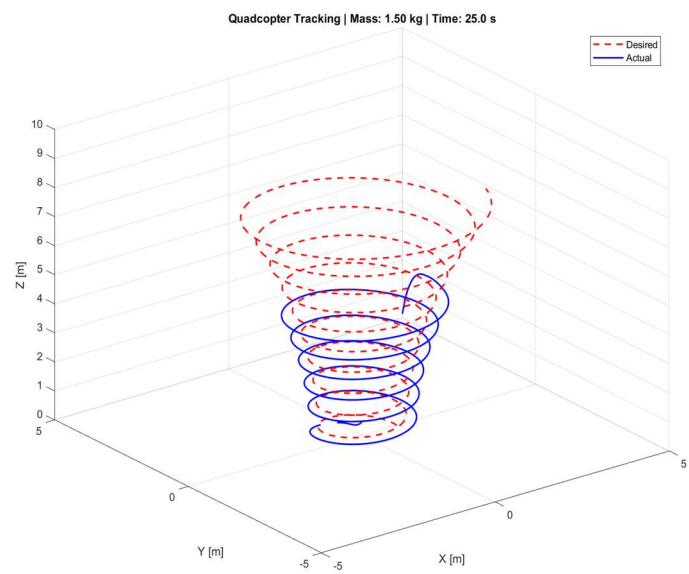


Fig. 15 3D trajectory tracking using the PSO-optimized PID controller under external disturbance.

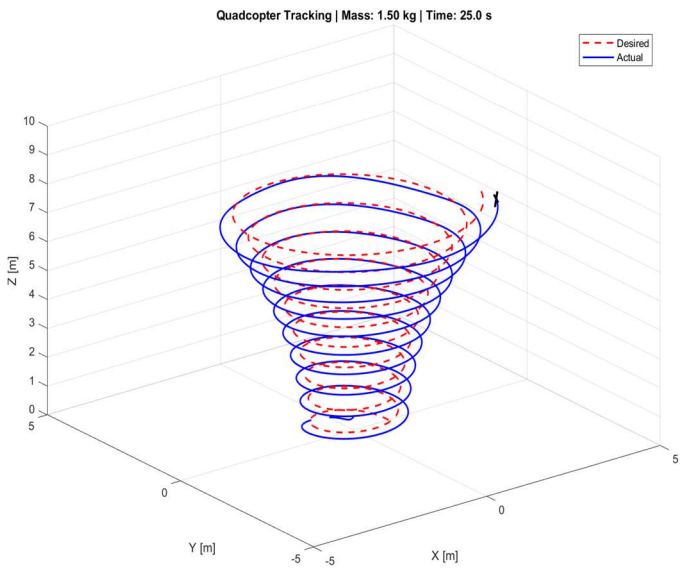


Fig. 16 3D trajectory tracking using the GA-optimized fuzzy self-tuning PID controller (no external disturbance).

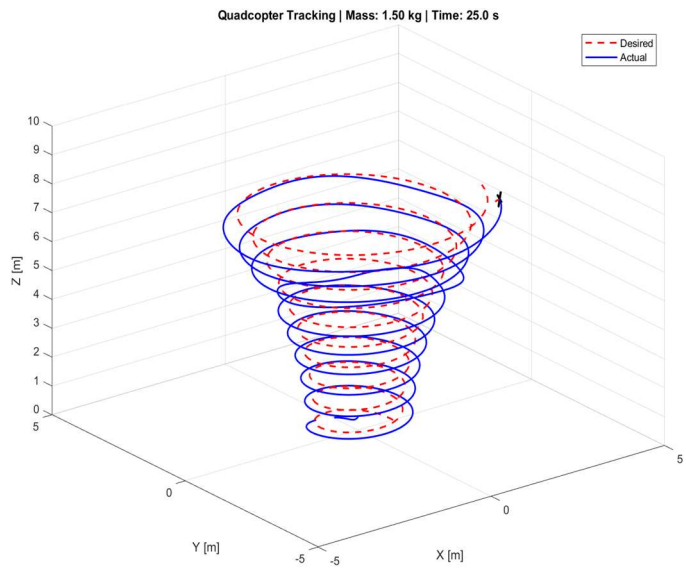


Fig. 17 3D trajectory tracking using the fuzzy self-tuning PID controller under external disturbance.

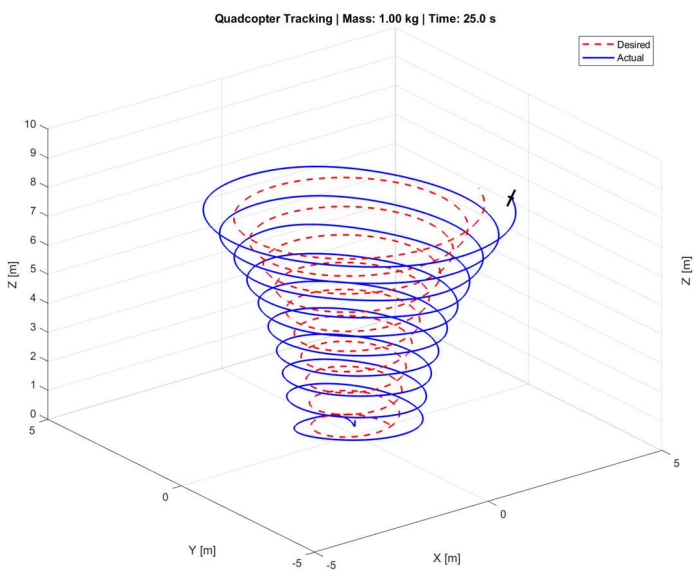


Fig. 18 3D trajectory tracking using the nominal PSO-PID controller under payload variation.

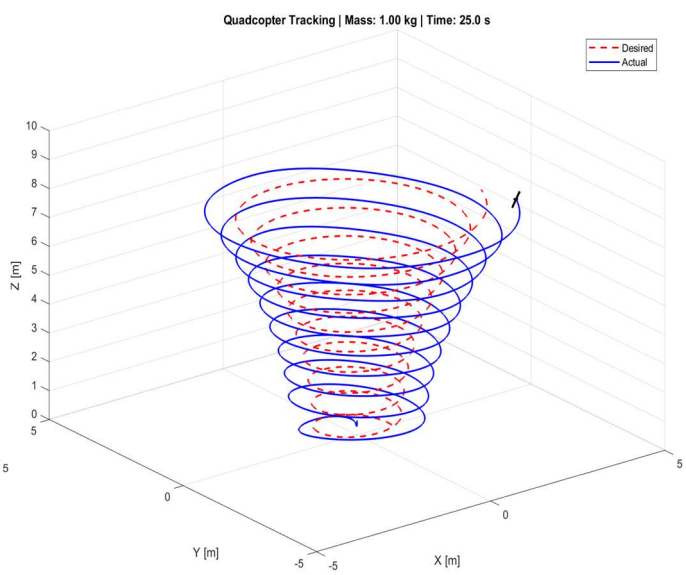


Fig. 19 3D trajectory tracking using the proposed controller under payload varation.

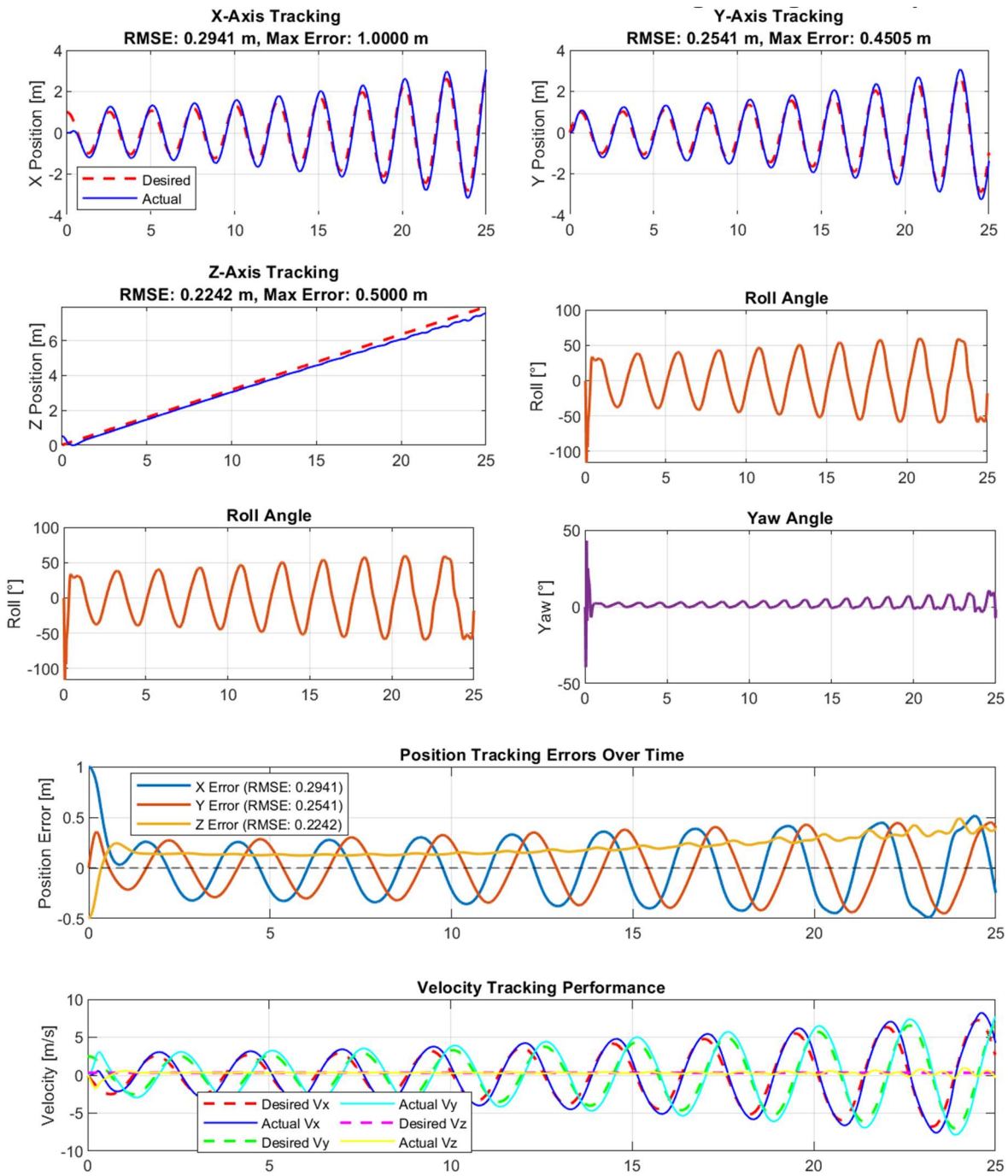


Fig. 20 Attitude and position responses of the PSO-optimized PID controller (no external disturbance).

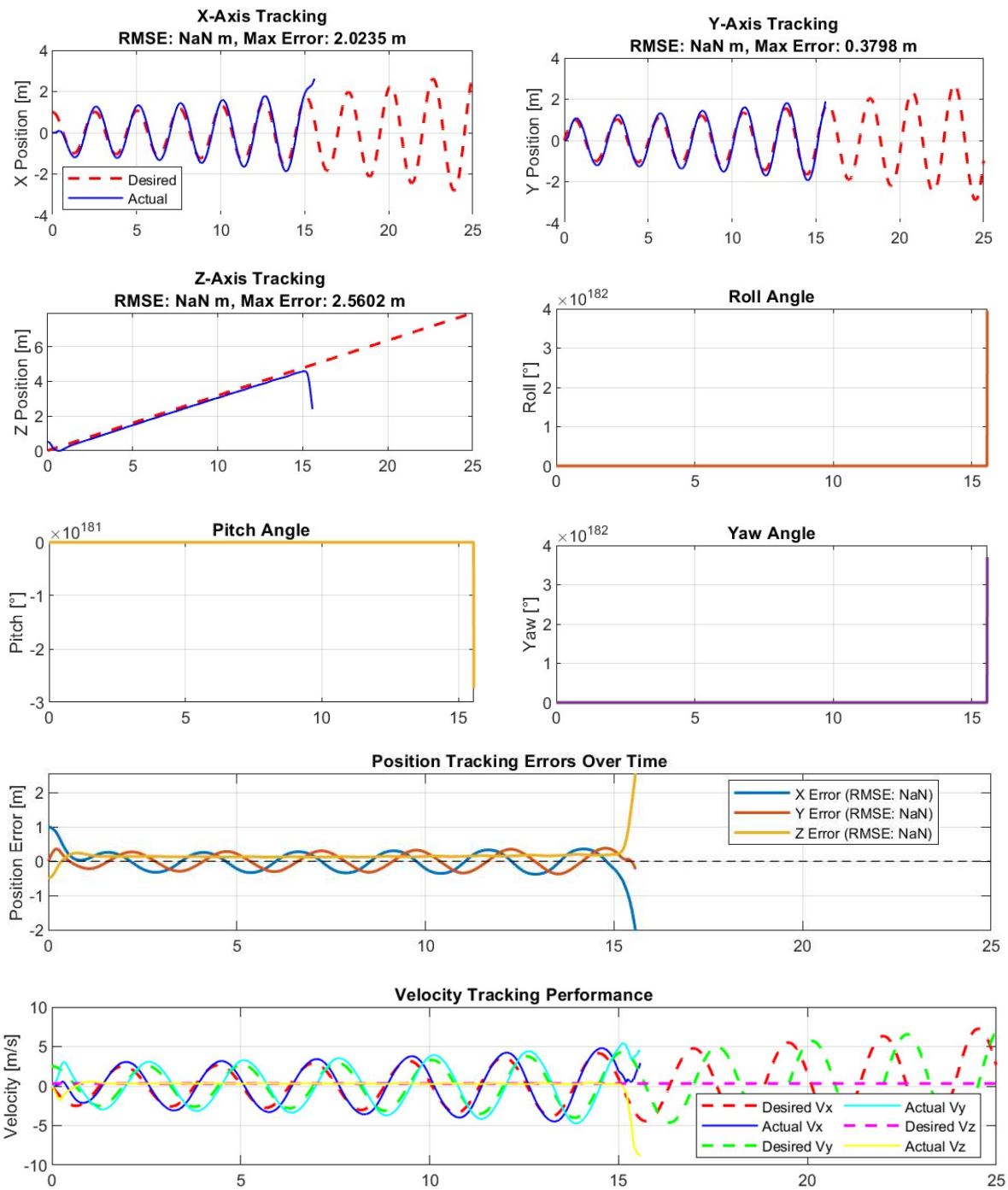


Fig. 21 Attitude and position responses of the PSO-optimized PID controller under external wind disturbance.

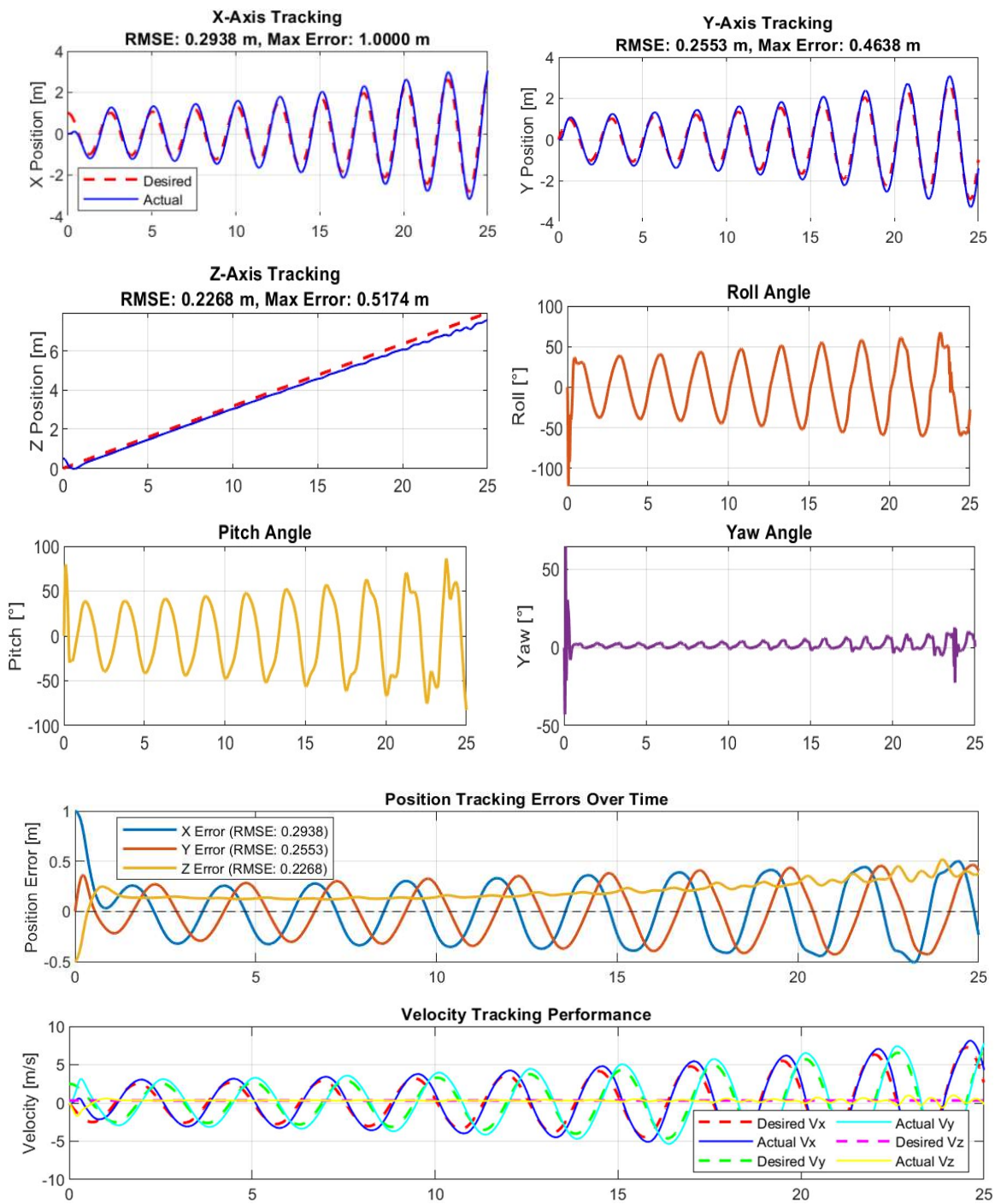


Fig. 22 Attitude and position responses of the fuzzy self-tuning PID controller (no external disturbance).

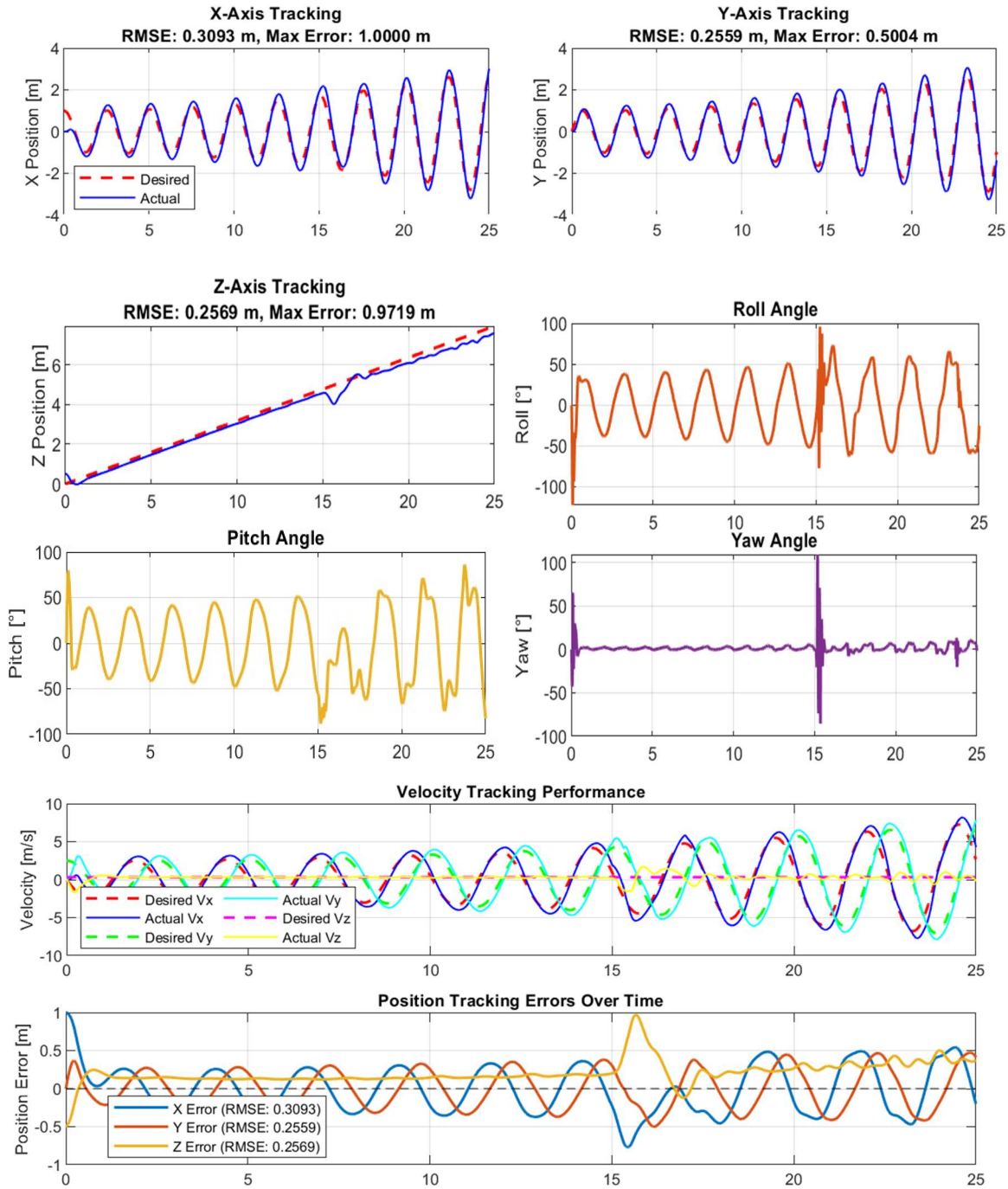


Fig. 23 Attitude and position responses of the fuzzy self-tuning PID controller under external wind disturbance.

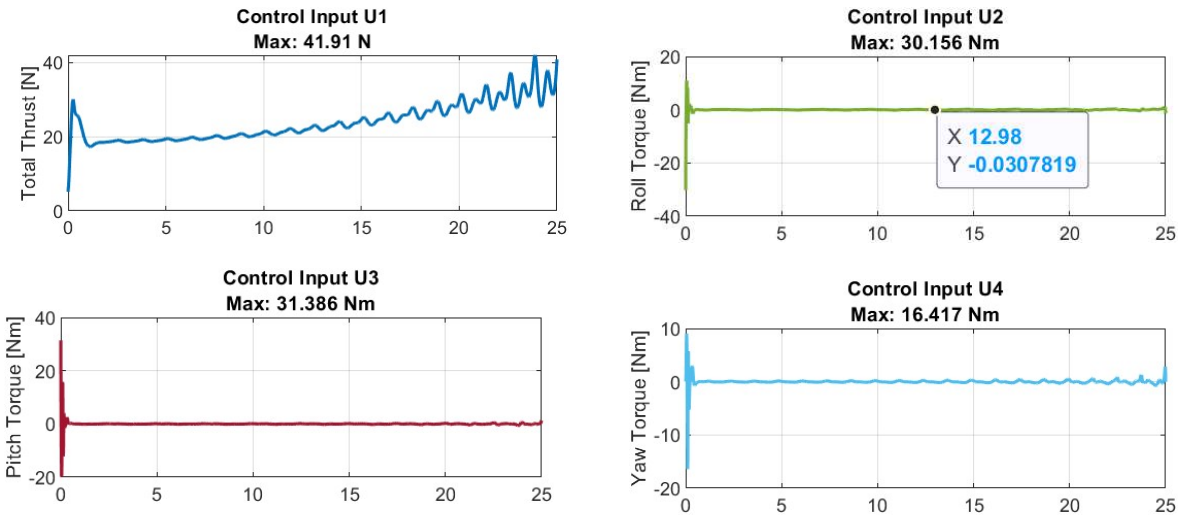


Fig. 24 Control inputs (total thrust and body torques) of the PSO-optimized PID controller (no external disturbance).

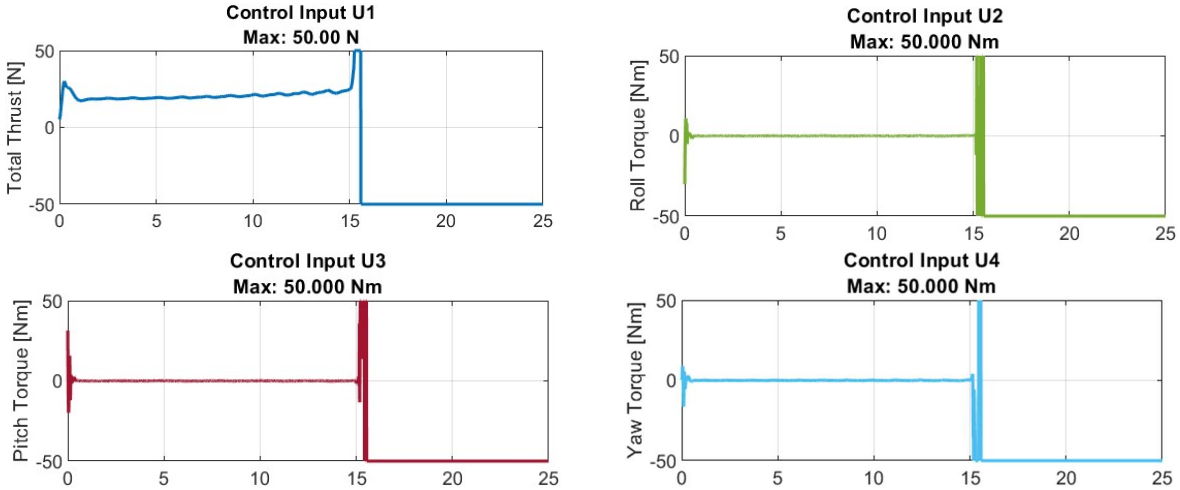


Fig. 25 Control inputs of the PSO-optimized PID controller under external disturbance.

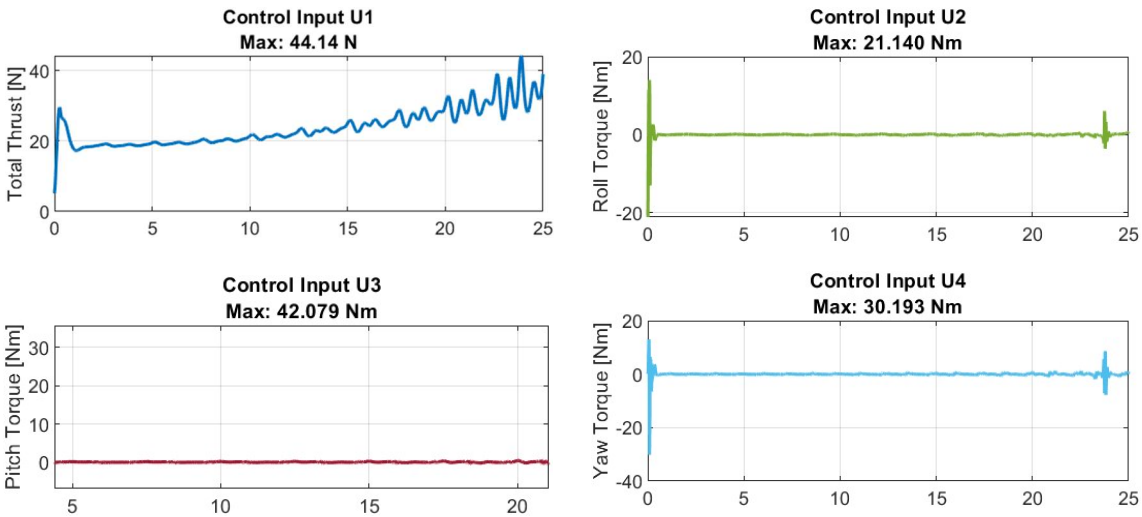


Fig. 26 Control inputs of the fuzzy self-tuning PID controller (no external disturbance).

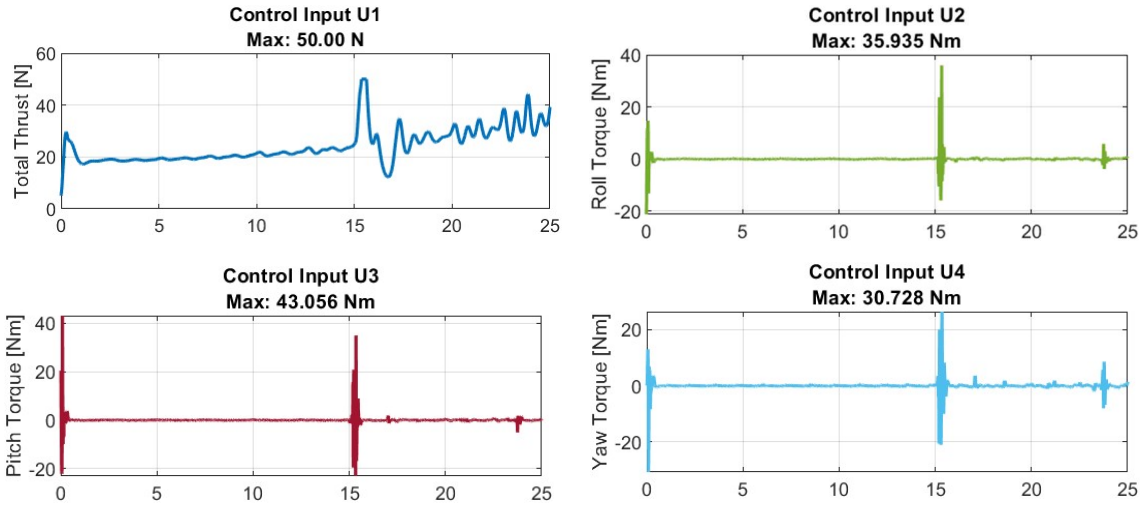


Fig. 27 Control inputs of the fuzzy self-tuning PID controller under external disturbance.

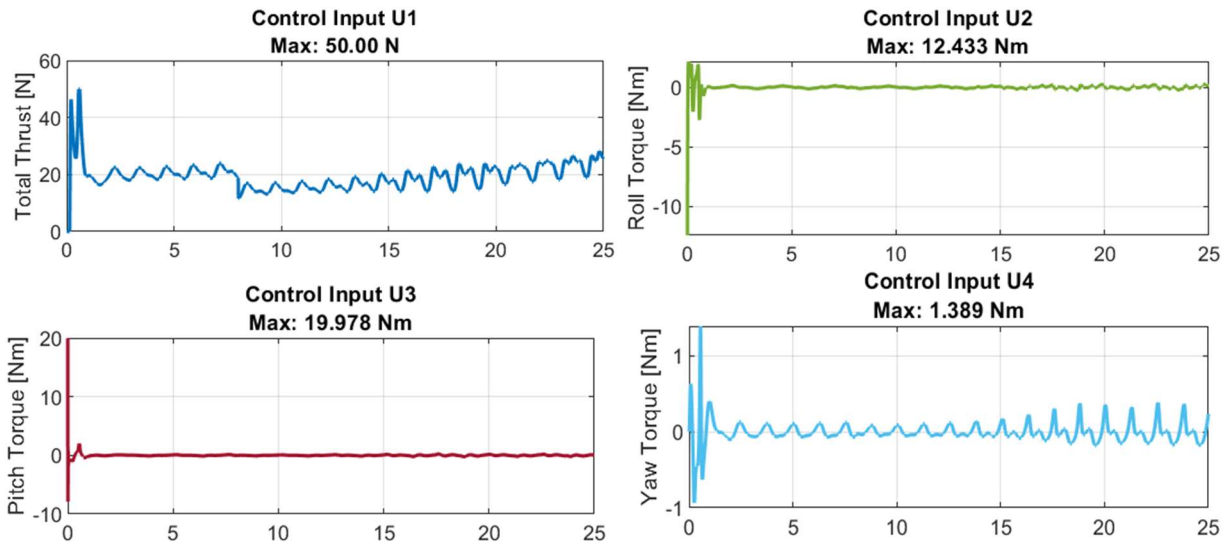


Fig. 28 Control inputs of the nominal (PSO-PID + LADRC) controller under payload variation.

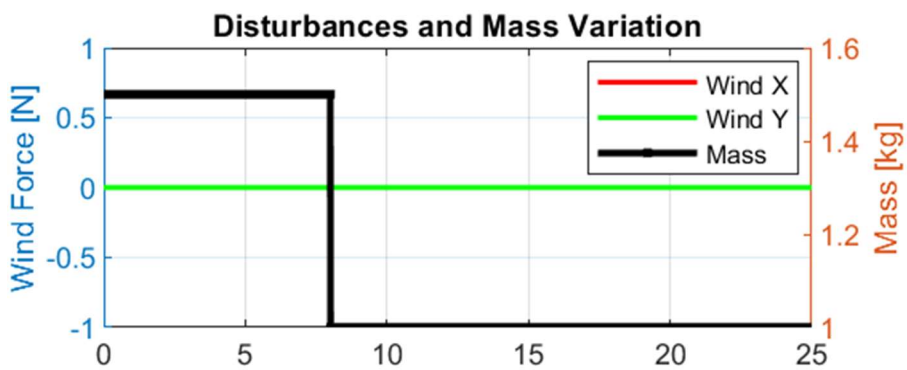


Fig. 29 Mass variation plot (the right-hand vertical axis denotes the total mass)

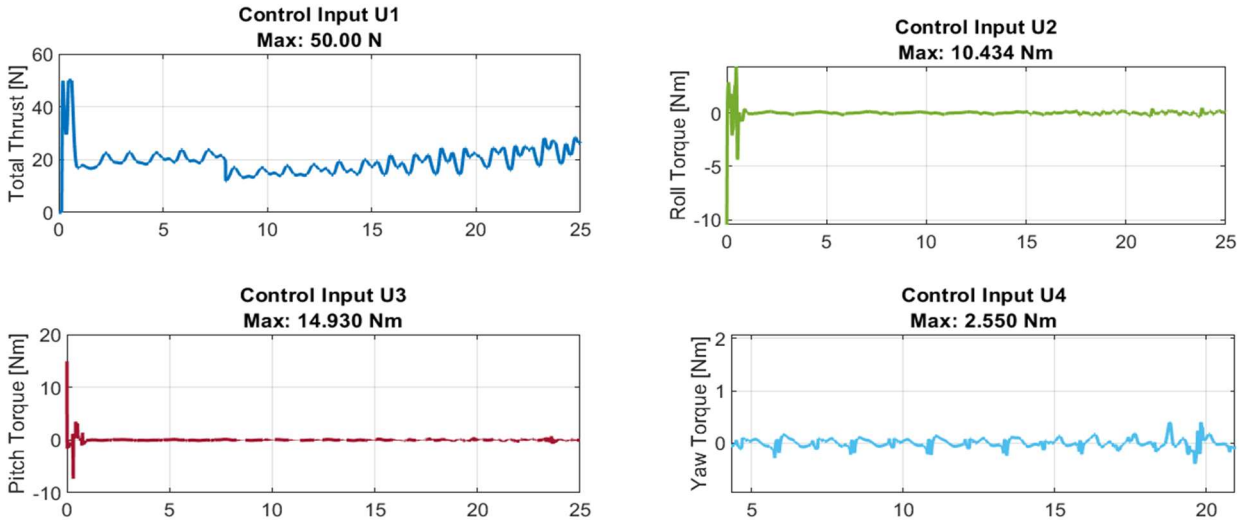


Fig. 30 Control inputs of the proposed controller under payload variations.

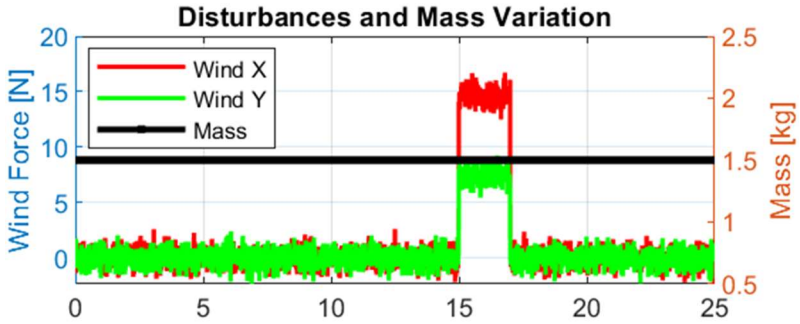


Fig. 31 Wind disturbance signals in x and y directions.

4.8 Overall Performance Summary Across All Scenarios (Wind Disturbance and Payload Variation)

The simulation results consistently demonstrate the superior performance of the proposed controller compared with the baseline PSO-tuned PID in both robust scenarios. Under wind disturbances, the baseline controller exhibits clear performance degradation and, in some cases, loss of tracking (as reflected by the unbounded ITAE values), whereas the proposed approach preserves stable closed-loop behavior with bounded errors and finite ITAE across all controlled states. This improvement is mainly attributed to the inner-loop fuzzy self-tuning, which strengthens attitude regulation during wind-induced coupling and enables faster disturbance attenuation. Similarly, under payload variation, the proposed controller maintains better tracking consistency after the abrupt mass change, achieving lower time-weighted tracking errors and smoother recovery than the fixed-gain baseline. This superiority is primarily due to the integration of LADRC in the altitude channel, whose main purpose is to reject the payload-induced thrust-to-weight mismatch by estimating and compensating the resulting total disturbance online. Overall, across both scenarios, the proposed controller achieves a better balance between tracking accuracy, robustness, and bounded control effort, confirming its suitability for practical operation under realistic environmental and operational uncertainties.

Table 13 ITAE performance indices of the compared controllers across all controlled states under nominal and disturbance scenarios.

Senario	Method	ITAE Index					
		x	y	z	v_x	v_y	v_z
Nominal	PSO-PID	86.2451	77.3024	76.9866	205.8352	194.2742	36.3764
	PSO-PID + Fuzzy-GA	86.4329	77.6081	77.6993	204.8051	194.0006	42.9770
Wind disturbance	PSO-PID	inf	inf	inf	inf	inf	inf
	PSO-PID + Fuzzy-GA	89.6612	77.7072	80.6530	208.2203	195.7843	78.1287
Payload variation	PSO-PID + LADRC	87.3659	184.7569	4.8654	211.6365	470.3046	22.9615

	PSO-PID + Fuzzy-GA + LADRC	86.8631	184.1567	4.6644	222.0227	469.0826	23.1966

5 CONCLUSION

This paper presented a multi-stage control strategy for quadcopter trajectory tracking based on a standard cascaded structure. First, Particle Swarm Optimization (PSO) was used to obtain a reliable baseline by tuning the PID controllers of the outer-loop horizontal position and the inner-loop attitude dynamics. Then, a supervisory fuzzy self-tuning mechanism was integrated within the inner loop to adapt the attitude PID gains online using the tracking error and its rate. To avoid arbitrary tuning of the adaptation strength, a Genetic Algorithm (GA) was implemented to optimize the fuzzy scaling factors offline. In addition, a Linear Active Disturbance Rejection Control (LADRC) scheme was assigned to the altitude channel, with the primary objective of handling payload variations that cause thrust-to-weight mismatch and degrade vertical tracking.

The simulation results confirmed that the PSO-tuned cascaded PID achieved stable performance in nominal conditions and provided a consistent baseline. However, under wind disturbances and payload changes, the fixed-gain baseline showed clear performance degradation. In contrast, the proposed controller maintained stable tracking and bounded control actions across all tested scenarios. Under wind disturbance, the improvement was mainly linked to the inner-loop fuzzy supervision, which strengthened attitude regulation and reduced coupling-induced tracking deviations. Under payload variation, the altitude LADRC improved recovery after mass change by estimating and compensating the induced total disturbance. Overall, the proposed approach achieved a better balance between tracking accuracy and robustness, and the ITAE-based comparison supported the superiority of the proposed strategy across the evaluated cases.

6 FUTURE WORK

Future work will focus on extending the validation beyond simulation toward more realistic implementation conditions. First, the controller will be tested through hardware-in-the-loop (HIL) and real-time embedded deployment to evaluate computation time, discretization effects, and sensor noise sensitivity. Second, the robustness analysis will be expanded to include actuator saturation, propeller dynamics, and time-varying wind fields, as well as uncertainty in inertial parameters. In addition, statistical model checking (SMC) will be investigated to provide probabilistic robustness guarantees under stochastic uncertainties (e.g., turbulent wind and measurement noise). This will be achieved by defining formal performance properties (e.g., bounded tracking error and settling-time constraints) and estimating their satisfaction probability

over large ensembles of randomized simulations, complementing the deterministic stress-test scenarios used in this work. Third, the LADRC altitude design can be further improved by adaptive tuning of the nominal gain and observer bandwidth to maintain performance across wider payload ranges. Finally, additional practical extensions will be explored, such as fault-tolerant control allocation, energy-aware optimization of the control effort, and formal stability/robustness analysis of the overall cascaded structure with fuzzy adaptation and disturbance estimation.

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